Adaptive Feature Recalibration Network for More Accurate Instance Segmentation in Remote Sensing

Xiao Yang and Ahmad Sufril Azlan Mohamed

***Abstract*—Accurate instance segmentation in remote sensing imagery is challenged by densely packed, arbitrarily oriented objects. Conventional horizontal-region-based approaches often suffer from feature misalignment, leading to degraded performance. To address this, we propose the Adaptive Feature Recalibration Network (AFRNet), a unified framework that iteratively refines rotated proposals in a coarse-to-fine manner. AFRNet introduces a Deformable Align Region Proposal Network (RPN) to generate geometry-aware oriented proposals, followed by a cascade of refinement stages utilizing Rotated Region of Interest (RoI) Align and stage-specific regressors. To handle angle periodicity and discontinuity, we model oriented boxes as 2-D Gaussian distributions, enabling a robust and continuous formulation for rotation-aware regression. Based on high-quality refined proposals, an orientation-aware mask head further enhances segmentation by producing precise, rotation-aligned instance masks. Furthermore, we integrate AFRNet with the Segment Anything Model (SAM) into a semisupervised framework that uses keypoints derived from oriented bounding boxes (OBBs) as prompts. Despite not requiring ground-truth masks, the resulting OBB-guided SAM delivers competitive segmentation results. Extensive experiments across six remote sensing benchmarks demonstrate that our rotation-aware design achieves state-of-the-art performance in both rotated object detection and instance segmentation, validating its effectiveness in complex aerial imaging scenarios.**

***Index Terms*—Adaptive feature recalibration, instance segmentation, remote sensing, rotated object detection.**

I. Introduction

COMPUTER vision has transformed remote sensing image analysis by enabling automated interpretation of high-resolution aerial and satellite data across a variety of fields. For instance, object detection techniques quickly identify and localize discrete objects in imagery, supporting applications such as urban planning and security monitoring [1]. Meanwhile, semantic segmentation achieves pixel-level classification to accurately delineate various land cover types, providing detailed insights for environmental monitoring and land cover analysis [2]. Despite these advances, many remote sensing challenges demand even finer granularity and greater precision. Instance segmentation bridges this gap by extending object detection with pixel-accurate delineation of individual objects, capturing their exact spatial boundaries and shapes. This integrated approach not only deepens the understanding of object sizes and interrelationships in complex scenes but also significantly enhances remote sensing applications [3].

Instance segmentation aims to detect every object in an image and delineate each with a precise pixel mask. Unlike semantic segmentation, which labels all objects of the same class as one, instance segmentation needs to distinguish adjacent or overlapping objects as separate entities. The dominant approach is the region-based two-stage paradigm: a network first proposes object candidates (typically as bounding boxes) and then refines each proposal to assign a class and predict a mask. For example, Faster R-CNN [4] introduced a Region Proposal Network (RPN) to efficiently generate box proposals. Mask R-CNN [5] builds on this by adding a small fully convolutional mask head: after the RPN proposes regions, a second-stage head classifies each box, refines its coordinates, and outputs a binary mask for the object. This explicit separation of detection and segmentation has proven very effective, yielding highly accurate instance masks. Since Mask R-CNN, many efforts have refined the two-stage framework to enhance mask quality. HRNet [6] maintains high-resolution features for fine-grained segmentation, while PANet [7] improves multiscale fusion via bottom-up pathways and adaptive pooling. GROI [8] improves region of interest (RoI) feature extraction by employing attention mechanisms to aggregate multilevel feature pyramid representations, and Mask Scoring R-CNN [9] calibrates mask confidence using predicted intersection over union (IoU). Cascade-based models like Cascade Mask R-CNN [10] and HTC [11] further improve results through multistage refinement, progressively enhancing localization and mask accuracy. Other paradigms have also been explored. One-stage models like YOLACT [12] and SOLO [13] skip explicit proposal generation, enabling faster inference at the expense of accuracy. Transformer-based approaches such as Mask2Former [14] unify detection and segmentation within an end-to-end query-driven framework, offering a clean and flexible alternative. Although existing instance segmentation methods achieve remarkable performance in natural image domains, their direct application to remote sensing imagery remains problematic. Most of these

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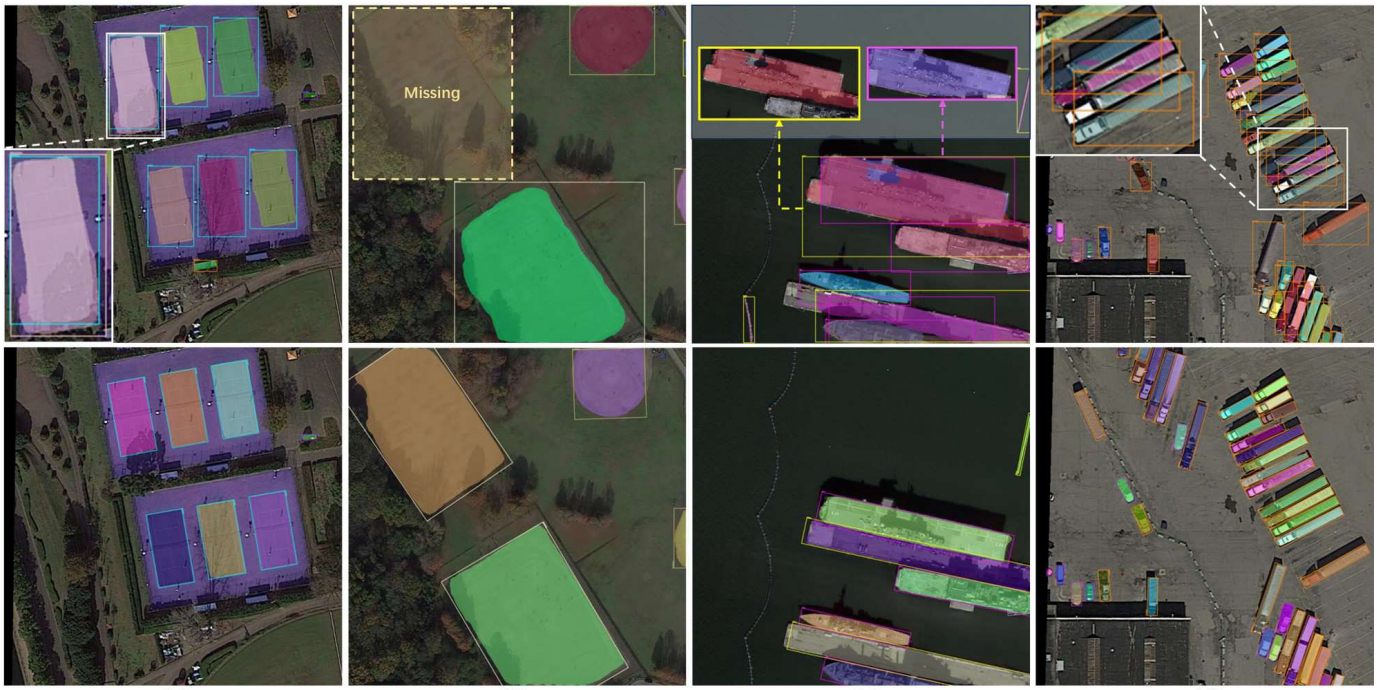


Fig. 1. Challenges in remote sensing instance segmentation and the effectiveness of our proposed AFRNet. The first row illustrates typical failure cases of conventional horizontal region-based methods in diverse remote sensing scenes. (a) Imprecise masks caused by excessive background inclusion, (b) missed detections due to interference from redundant features, (c) severe mask contamination in densely populated harbor scenes, and (d) overlapping boxes causing missed instances via NMS suppression. The second row presents the corresponding outputs from our AFRNet, which leverages rotated proposals and orientation-aware feature alignment to produce more accurate and complete segmentation results under complex object layouts and orientations.

approaches rely on horizontal bounding box (HBB) detection, which proves inadequate in scenes with densely distributed and arbitrarily oriented objects. Unlike natural images captured from a fixed ground-level perspective, remote sensing imagery is acquired from an overhead viewpoint, leading to pronounced orientation variations. Objects with elongated aspect ratios such as ships, harbors, and vehicles are common in aerial scenes, and when they appear in nonaxis-aligned orientations, HBBs inevitably cover excessive background areas. Region-based instance segmentation frameworks like Mask R-CNN inherit this limitation, as their mask prediction heads operate on horizontal proposals. In remote sensing scenarios, where objects are both densely packed and arbitrarily oriented, this design causes substantial misalignment between proposals and true object boundaries, ultimately degrading segmentation accuracy [15].

As shown in the first row of Fig. 1, conventional instance segmentation models often struggle with the unique challenges of remote sensing imagery—such as diverse object orientations, extreme aspect ratios, and densely packed scenes—which collectively contribute to notable performance degradation. In Fig. 1(a), horizontal proposals often include large areas of background, resulting in imprecise masks filled with irrelevant pixels and leading to incorrect recall. In Fig. 1(b), the excessive background not only degrades segmentation accuracy but can also cause the object to be completely missed. Moreover, in environments like harbors or parking lots, instances often appear close together with overlapping regions. In Fig. 1(c), closely packed ships and harbor infrastructure interfere with each other's segmentation masks, causing significant mutual pixel contamination. In Fig. 1(d), vehicles in a crowded parking lot result in overlapping horizontal proposals. These overlaps frequently trigger nonmaximum suppression (NMS) to mistakenly discard valid detections, leading to missed instances.

Given the unique challenges in remote sensing imagery, there is an urgent need for instance segmentation models specifically tailored to this domain. Recent progress in rotated object detection has introduced innovative solutions, including advanced representations of oriented bounding boxes (OBBs) and the integration of angle prediction branches, which significantly improve the modeling of object orientation. These methods overcome the limitations of conventional HBBs and enhance the geometric representation of ground objects, offering strong potential for downstream instance segmentation tasks [16].

Despite their advantages in object detection, extending OBB-based approaches to high-quality instance segmentation introduces several new challenges. First, simply appending an angle regression branch to a detection network without orientation-aware feature alignment can result in significant feature misalignment, particularly in crowded scenes where extracted features are easily contaminated by background or neighboring objects [17]. Second, angle regression is challenged by inherent periodicity and discontinuity. For instance, under the long-edge definition, the rotation angle θ is restricted

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to the range $[-90°, 90°)$; while $-89°$ and $89°$ are geometrically close, they are numerically far apart, which can lead to unstable training and hinder convergence [18]. Moreover, most rotated detectors are evaluated under relatively loose Rotated IoU (RIoU) thresholds (e.g., 0.5), meaning that predictions with substantially different orientations may still be considered positive. As a result, the average precision (AP) tends to drop sharply when stricter RIoU thresholds are applied. Such inaccurate orientation predictions can severely degrade the quality of subsequent instance segmentation, where precise localization and alignment are essential.

To overcome these limitations, we introduce the Adaptive Feature Recalibration Network (AFRNet), a unified framework that dynamically adjusts receptive fields to achieve coarse-to-fine feature alignment for accurate rotated object detection and instance segmentation. AFRNet attains competitive detection performance even under stricter RIoU thresholds. Building on these refined rotated proposals, AFRNet is further extended to instance segmentation through an orientation-aware mask head that exploits orientation-consistent features to produce high-quality masks. As illustrated in the second row of Fig. 1, AFRNet delivers more precise segmentation of arbitrarily oriented objects, particularly in densely packed scenes.

Moreover, as datasets with OBB annotations for rotated object detection are more widely available than those with pixel-level mask annotations for instance segmentation, we propose a semisupervised extension by integrating AFRNet with the Segment Anything Model (SAM) [19]. Our OBB-guided SAM strategy leverages keypoints derived from OBBs as geometric prompts, enabling high-quality mask generation without the need for ground-truth mask supervision.

The contributions of our work are as follows.

1) We propose AFRNet, which integrates a Deformable Align RPN (DARPN) to generate orientation-aware proposals via shared modulated deformable convolutions, and a cascade of refinement stages that leverage Rotated RoI (RRoI) Align and Gaussian modeling to enhance feature alignment and stabilize angle regression. Together, these components enable robust and accurate detection of arbitrarily oriented and densely packed objects, providing a strong foundation for downstream instance segmentation tasks. Extensive experiments on four remote sensing rotated object detection datasets demonstrate that AFRNet achieves state-of-the-art detection accuracy.
2) We extend AFRNet to instance segmentation by designing an orientation-aware mask head that conditions segmentation on rotation-aligned features, enabling high-quality instance masks that closely match the geometry of nonaxis-aligned objects in complex aerial scenes. Experiments on two remote sensing instance segmentation benchmarks show that our method attains state-of-the-art mask quality.
3) We introduce an OBB-guided SAM strategy that integrates AFRNet with SAM to enable semisupervised instance segmentation using only box-level annotations. By leveraging keypoints derived from OBBs, this approach achieves competitive mask quality without relying on ground-truth mask supervision. Evaluations on two remote sensing instance segmentation datasets confirm its superior performance, and additional visualizations on four rotated object detection datasets further demonstrate its robustness.

II. Related Works

A. Rotated Object Detection Methods

Rotated object detection, like general object detection, can be categorized into two-stage and one-stage approaches. Among two-stage methods, Rotated Faster R-CNN [4], [20] marks a significant advancement in rotated object detection technology by integrating orientation regression. The model introduces an angle parameter to represent orientation information, along with an additional output channel specifically designed for regressing the orientation of each object. RRPN [17], [21] enhances accuracy by incorporating a larger set of rotated anchors across six orientations and performing angle parameter regression during the RPN stage. It also introduces RRoI Pooling operation to extract features. To minimize the reliance on numerous rotated anchors, RoI Transformer [22] introduces an additional stage called RRoI Learner, which performs coarse-grained learning of orientation information before passing it to R-CNN for more refined regression of orientation details. Furthermore, it adopts RRoI Align operation to enhance spatial feature alignment for arbitrarily oriented objects. Beyond rotated anchor generation strategies, another line of research focuses on refining the representation of rotated objects. Gliding Vertex [23] introduces an innovative regression paradigm that slides the vertices of an HBB along its edges, thereby reconstructing an OBB without relying on explicit angle regression. Oriented R-CNN [15] develops an Oriented RPN with a midpoint offset representation, which derives oriented boxes from the enclosing horizontal boxes and the offsets of selected midpoints, thereby preserving the horizontal regression mechanism and leading to more stable training with robust convergence. QPDet [24] proposes the quadrant point representation, which parameterizes an oriented rectangle with the box center, enclosing circumcircle, and symmetrical offsets, naturally eliminating boundary discontinuity in regression.

As a representative one-stage rotated object detector, Rotated RetinaNet [20], [25] replaces the standard bounding box regression of RetinaNet with OBB regression, performing both classification and regression in a single stage using focal loss. While one-stage detectors are computationally efficient, they often compromise precision compared to two-stage detectors, particularly in rotated object detection, where feature misalignment is a frequent challenge. To address this, various algorithms have been developed to incorporate lightweight refinement stages that enhance feature alignment for rotated objects. R3Det [26] introduces a feature refinement module that incorporates pixelwise feature interpolation to integrate the positional information of refined bounding boxes into corresponding feature points, enabling feature reconstruction and alignment. S2ANet [27] addresses feature misalignment through its feature alignment module, which employs alignment convolution to infer offsets that adjust the feature grid

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to align with refined rotated anchors. DCFL [28] dynamically models prior knowledge, label assignment, and object representation. It adopts a dual strategy for label assignment, integrating coarse prior matching with fine-grained posterior constraints to ensure balanced and effective supervision across diverse instances. Anchor-free detectors, such as rotated FCOS [20], [29], eliminate reliance on predefined anchors and are inherently more flexible in handling objects with arbitrary orientations. However, directly regressing both the position and angle of rotated bounding boxes introduces additional optimization challenges. Without anchor-based priors, training stability can be compromised, necessitating carefully designed loss functions and tailored training strategies [30].

Despite substantial progress in rotated object detection, most existing methods remain primarily optimized for AP_{50} and suffer a pronounced performance drop under stricter RIoU thresholds, highlighting limited localization precision. To address this challenge, we propose AFRNet, a unified framework that augments the oriented RPN with deformable alignment to generate high-quality oriented proposals and incorporates a cascade refinement module that integrates RRoI Align with Gaussian-based regression. Each cascade stage is trained with progressively stricter RIoU thresholds, enabling incremental refinement of localization accuracy. This joint design facilitates orientation-aligned feature extraction, ensures stable angle regression, and progressively enhances localization. Through these innovations, AFRNet achieves robust and precise detection of arbitrarily oriented objects, surpassing existing methods particularly under high RIoU metrics, while establishing a solid foundation for downstream tasks such as instance segmentation.

B. Instance Segmentation Methods

Instance segmentation methods can be broadly categorized based on their underlying object detection framework. Two-stage methods adopt a sequential pipeline, first generating region proposals and then refining them to produce precise instance masks. Among these methods, Mask R-CNN [5] serves as a benchmark, extending Faster R-CNN [4] with a parallel mask branch that enables simultaneous object detection and high-quality instance mask generation. HRNet [6] introduces a high-resolution representation framework that preserves spatial details throughout the network, yielding notable improvements in fine-grained segmentation, especially for densely packed or small-scale objects. PANet [7] enhances the feature hierarchy through bottom-up path augmentation, which strengthens localization signals in lower layers, and adaptive feature pooling, which aggregates multilevel features for more effective proposal generation. GRoI [8] proposes a unified RoI extraction layer that treats all feature pyramid levels as equally informative, extracting fixed-size features from each scale independently and aggregating them with attention to form a context-enriched representation. Mask Scoring R-CNN [9] further improves Mask R-CNN by introducing a mask IoU prediction branch that aligns mask confidence with actual quality, thereby yielding more reliable predictions. Cascade-based architectures also advance segmentation: Cascade Mask R-CNN [10] refines masks through progressively increasing IoU thresholds, HTC [11] interweaves detection and segmentation in an multistage framework, and SCNet [31] enforces consistency between training and inference IoU distributions, enhancing both robustness and accuracy.

One-stage instance segmentation methods streamline the pipeline by predicting object masks in a single forward pass. YOLACT [12] adopts a dual-branch design that decouples mask generation from instance localization by producing a set of shared prototype masks and predicting per-instance mask coefficients, which are linearly combined to form final masks, thereby eliminating the need for per-RoI feature extraction. SOLO [13] reformulates instance segmentation as a location-based categorization task, partitioning the image into grids and assigning each pixel to an instance category based on its spatial position and scale. SOLOv2 [32] further enhances this framework with dynamic convolutional heads and Matrix NMS, achieving a favorable balance between accuracy and efficiency. CondInst [33] employs dynamic instance-aware networks conditioned on each instance, enabling fully convolutional mask prediction without RoI operations and allowing for a compact yet effective mask head.

Transformer-based models have recently achieved remarkable success in computer vision and are increasingly being adopted for remote sensing segmentation. However, most existing transformer-based methods in this field primarily address semantic segmentation [34], whereas instance segmentation has received comparatively limited attention [35]. Mask2Former [14], an extension of MaskFormer [36], uses a transformer decoder with masked attention, constraining cross-attention to predicted mask regions and providing a unified framework for semantic, instance, and panoptic segmentation. These transformer-based approaches offer elegant, end-to-end solutions with strong generalization capabilities.

To reduce the high annotation cost of pixel-level masks, semisupervised methods have emerged. BoxInst [37] learns instance masks from bounding boxes using a projection-based surrogate loss and a color-similarity pairwise loss, eliminating the need for mask annotations. DiscoBox [38] jointly learns instance segmentation and semantic correspondence under box supervision via a structured teacher model, generating pseudo masks and correspondences for self-ensembling training. Box2Mask [39] iteratively evolves level-set curves guided by image features and a local pixel-affinity module, optimizing masks within bounding boxes using an instance-aware decoder and box-level matching. These box-supervised techniques demonstrate that competitive instance segmentation is achievable with significantly reduced labeling effort.

Prompt-driven segmentation models have further advanced the field. SAM [19], trained on over one billion masks, enables category-agnostic segmentation with minimal user input, such as points, boxes, or masks, while SAM2 [40] extends this capability to video. In remote sensing, specialized adaptations include SAMRS [41], which converts bounding box annotations into masks; PMHO [42], which employs progressive multiscale hierarchical prompting to refine object delineation; UrbanSAM [43], which uses invariance-inspired adapters for complex urban scenes; and PointSAM [44], which leverages point-level prompts to enhance segmentation performance.

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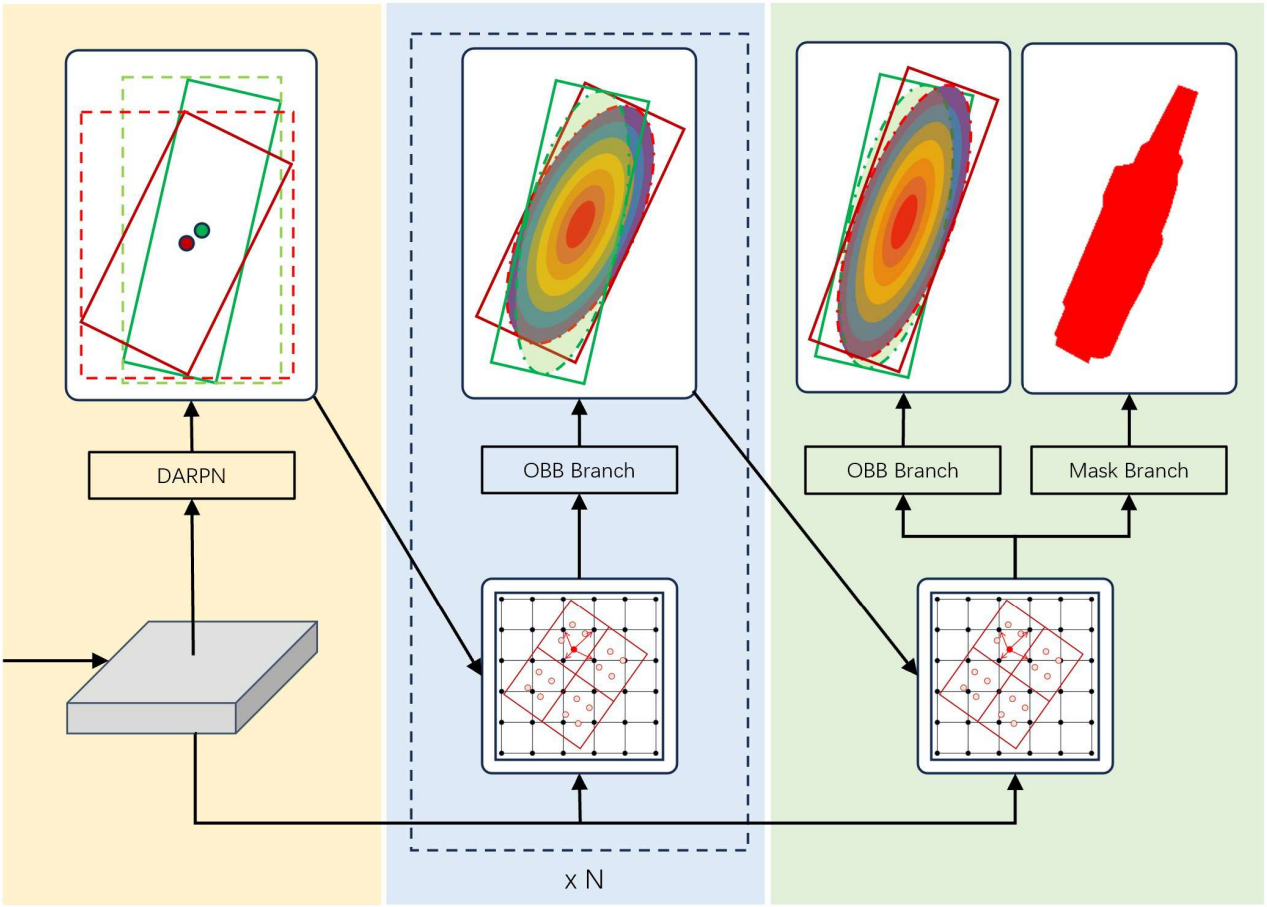


Fig. 2. Overall framework of the AFRNet. AFRNet follows a coarse-to-fine cascade strategy. It starts with oriented proposals generated by the DARPN module and refines them through multiple stages using RRoI Align and stage-specific heads. Gaussian modeling is applied at each stage to improve box regression. Finally, an orientation-aware mask head produces precise, rotation-aligned instance masks based on the refined proposals.

In remote sensing imagery, instance segmentation is challenged by large variations in object orientations and densely clustered scenes. Conventional approaches that rely on HBBs as proposals or prompts often prove inadequate, as they lack the orientation specificity required for precise segmentation. To address this, we extend AFRNet with an orientation-aware mask head that benefits from cascade-refined, rotation-aligned features, enabling the generation of high-quality masks that faithfully capture the geometry of arbitrarily oriented objects. Central to this design is the use of high-quality rotated proposals that remain reliable even under the stringent localization demands of strict RIoU thresholds, where minor misalignments can otherwise lead to substantial mask degradation. In addition, we propose an OBB-guided SAM strategy, where keypoints derived from OBBs serve as geometric prompts, achieving competitive mask generation even without ground-truth supervision. Collectively, these innovations contribute to advancing both fully supervised and semisupervised learning for remote sensing instance segmentation.

III. Methods

A. Adaptive Feature Recalibration Network

Our AFRNet operates on multiscale feature maps generated by the Feature Pyramid Network (FPN) [45] in a coarse-to-fine manner. As illustrated in Fig. 2, it begins with a DARPN module that generates oriented object proposals. DARPN employs modulated deformable convolutions [46] to learn spatial sampling offsets, enabling the receptive field to adapt to object geometry. This results in high-quality oriented proposals represented using a midpoint offset formulation [15]. Subsequently, a cascade of refinement stages progressively improves these proposals. Each stage uses RRoI Align [22] to extract orientation-aligned features and stage-specific regressors to update bounding boxes. This coarse-to-fine strategy enhances localization precision and handles complex object shapes more effectively. To further improve orientation regression, AFRNet models OBBs as 2-D Gaussian distributions, mitigating issues related to angle periodicity and discontinuity. Finally, an orientation-aware mask head predicts instance segmentation masks for each refined proposal. By conditioning mask generation on orientation-aligned features, the resulting masks more accurately capture object boundaries, especially for rotated or densely packed instances.

1) Deformable Align RPN: As shown in Fig. 3, the DARPN consists of two parallel branches: a classification branch for objectness prediction and a regression branch for generating oriented proposals. Both branches begin with a dilated convolution to capture broader contextual information. To model geometric variations, the regression branch incorporates a modulated deformable convolution to predict spatial offsets

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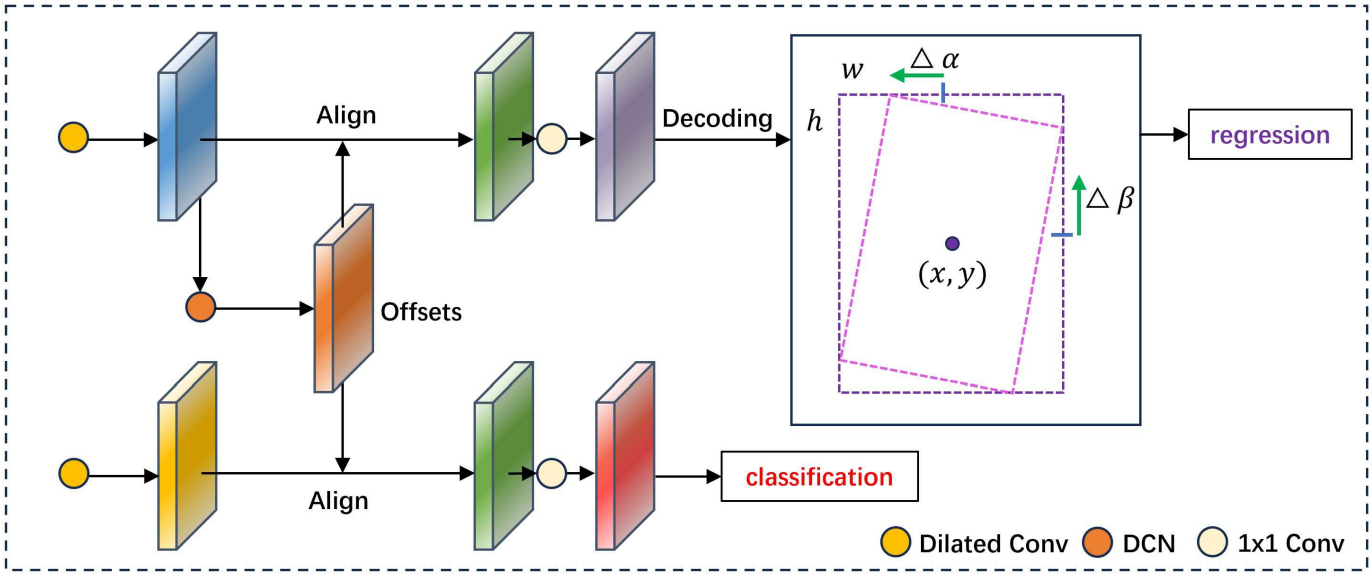


Fig. 3. Architecture of the DARPN. DARPN features parallel classification and regression branches, both initialized by dilated convolutions. The regression branch predicts spatial offsets and modulation scalars via a modulated deformable convolution, which are shared with the classification branch to ensure consistent, geometry-aware sampling. The regression outputs six-parameter offsets, decoded into midpoint-based oriented proposals and normalized to rotated rectangles, enabling accurate proposal generation under complex object geometries.

and modulation scalars. These predicted offsets and scalars are then shared with the classification branch to guide its deformable convolution, ensuring consistent sampling patterns across both branches. Finally, the classification branch outputs objectness scores, while the regression branch produces six-parameter offsets, which are decoded into a midpoint offset representation for oriented proposals.

Let $\mathbf{x}_{\text{in}}$ denote the input feature map. The regression branch applies a modulated deformable convolution module $\mathcal{D}_{\text{reg}}(\cdot)$, yielding three outputs

$$\left[\mathbf{x}_{\text{reg}},\, \Delta p,\, m\right] = \mathcal{D}_{\text{reg}}\left(\mathbf{x}_{\text{in}}\right) \tag{1}$$

where $\mathbf{x}_{\text{reg}}$ is the output feature map after deformable convolution, Δp represents the learned spatial offsets at each sampling location, and m contains the modulation scalars after sigmoid activation.

The offsets Δp and preactivation modulation scalars $\tilde{m}$ are generated by an auxiliary convolutional layer $\mathcal{C}_{\text{offset}}(\cdot)$, which shares the same kernel size, stride, padding, and dilation as the deformable convolution. Its output has $3K$ channels: the first $2K$ channels encode the offsets Δp, and the remaining K channels are passed through a sigmoid function to obtain the modulation scalars

$$\left[\Delta p,\, \tilde{m}\right] = \mathcal{C}_{\text{offset}}\left(\mathbf{x}_{\text{in}}\right), \quad m = \sigma\left(\tilde{m}\right) \tag{2}$$

where $\sigma(\cdot)$ denotes the elementwise sigmoid function mapping values to $[0, 1]$.

The modulated deformable convolution augments a regular sampling grid $\{p_k\}_{k=1}^{K}$ centered at each spatial position with learned offsets Δp_k, yielding deformed locations $p + p_k + \Delta p_k$. Each sampled value is modulated by both the learned scalar m_k and a trainable weight $\mathbf{w}_k$. The output at position p is computed as follows:

$$\mathbf{x}_{\text{out}}\left(p\right) = \sum_{k=1}^{K} \mathbf{w}_k \cdot m_k \cdot \mathbf{x}_{\text{in}}\left(p + p_k + \Delta p_k\right) \tag{3}$$

where the modulation scalars act as adaptive weights for individual sampling locations. Without them, all sampled features contribute equally, which may introduce noise from irrelevant regions such as background areas. By incorporating modulation scalars, the network can dynamically suppress or emphasize specific samples, thereby improving robustness and enhancing the representation of fine-grained structures.

To ensure geometric consistency between the two branches, DARPN employs a parameter-sharing strategy: the offsets Δp and modulation scalars m, predicted by the regression branch, are reused in the classification branch to perform deformable sampling with identical spatial patterns. Specifically, the classification branch applies modulated deformable convolution in the same manner as defined in (3), using the deformation parameters generated by the regression branch

$$\mathbf{x}_{\text{cls}} = \mathcal{D}_{\text{cls}}\left(\mathbf{x}_{\text{in}}, \Delta p, m\right). \tag{4}$$

This cross-branch alignment ensures that both branches sample from precisely the same spatial locations, thereby enhancing task coherence across the network. By relying on shared visual evidence, the classification and regression branches avoid the spatial inconsistencies typically introduced by independently learned sampling patterns. The offsets learned by the regression branch guide the classification branch toward semantically rich regions—such as object boundaries—without incurring any additional computational overhead. Furthermore, since the deformation parameters are jointly optimized through gradients from both classification and regression losses, the learning process becomes more

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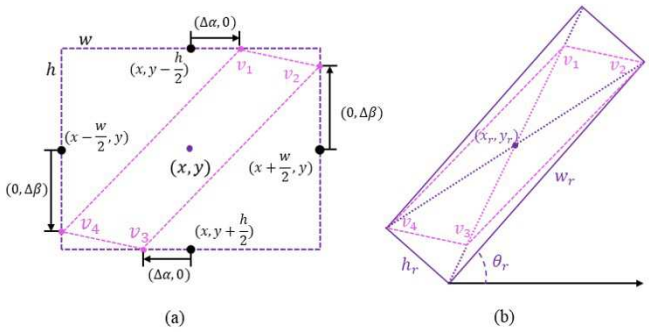


Fig. 4. Illustration of the transformation from the midpoint offset representation to a rotated rectangle. (a) Parallelogram constructed from midpoint offsets and (b) diagonal-based normalization to yield a rotated rectangle.

stable and consistent. This tightly coupled design facilitates accurate, orientation-aware region proposals, enabling DARPN to effectively handle complex object geometries and spatial deformations.

Instead of regressing the rotation angle θ directly, we adopt a midpoint offset representation $(x, y, w, h, \Delta\alpha, \Delta\beta)$ as an intermediate encoding of the oriented proposal. Each anchor is defined by the tuple (x_a, y_a, w_a, h_a), where (x_a, y_a) denotes the center coordinates, and w_a and h_a represent the anchor's width and height, respectively. The regression branch outputs relative offsets $(\delta_x, \delta_y, \delta_w, \delta_h, \delta_\alpha, \delta_\beta)$, which are decoded as follows:

$$\begin{cases} x = \delta_x \cdot w_a + x_a \\ y = \delta_y \cdot h_a + y_a \\ w = w_a \cdot \exp(\delta_w) \\ h = h_a \cdot \exp(\delta_h) \\ \Delta\alpha = \delta_\alpha \cdot w \\ \Delta\beta = \delta_\beta \cdot h \end{cases} \tag{5}$$

where (x, y, w, h) defines the external rectangle, while $\Delta\alpha$ and $\Delta\beta$ are offsets to the midpoints of its top and right edges. This six-parameter representation defines an initial oriented proposal in the form of a parallelogram.

To convert the parallelogram into a valid rectangle, the four vertices (v_1, v_2, v_3, v_4) are first reconstructed by shifting the midpoints of the external rectangle's edges according to the decoded offsets $(\Delta\alpha, \Delta\beta)$, as illustrated in Fig. 4(a). Next, the two diagonals of the parallelogram are measured with respect to the box center. As shown in Fig. 4(b), the shorter diagonal is extended to match the longer one, thereby enforcing the geometric constraint that the diagonals of a rectangle are equal in length, which in turn guarantees orthogonality of its edges. The resulting rectangle is parameterized as $(x_r, y_r, w_r, h_r, \theta_r)$, where the center (x_r, y_r) is inherited from the external rectangle, the width w_r and height h_r correspond to the lengths of the longer and shorter edges, respectively, and the rotation angle θ_r is determined by the orientation of the longer edge with respect to the x-axis.

2) Cascade Gaussian-Based RCNN: While DARPN provides accurate initial oriented proposals, further refinement is crucial for high-precision modeling of rotated objects. To this end, AFRNet incorporates a multistage cascade framework, where a sequence of detection heads progressively refines the proposals. Each stage in the cascade is trained with an increasing IoU threshold, allowing the model to focus on increasingly precise localization. By feeding the output of one stage into the next, the framework adaptively resamples hard examples and incrementally improves detection quality. This cascade design ensures that each stage benefits from a richer set of positive samples, thereby narrowing the performance gap between low- and high-quality proposals. The same refinement process is mirrored during inference, ensuring consistency between training and testing.

Formally, let $\mathbf{x}$ denote the backbone feature maps, and $\mathbf{b}_{n-1}$ be the OBBs from the $(n-1)$th stage (with $\mathbf{b}_0$ obtained from DARPN). The nth stage performs the following operations:

$$\mathbf{x}_n^{\text{box}} \leftarrow \mathcal{P}(\mathbf{x}, \mathbf{b}_{n-1}), \quad \mathbf{b}_n \leftarrow \mathcal{B}_n\left(\mathbf{x}_n^{\text{box}}\right) \tag{6}$$

where $\mathcal{P}(\cdot)$ denotes RRoI Align, which extracts region-aligned features $\mathbf{x}_n^{\text{box}}$ based on the oriented proposals $\mathbf{b}_{n-1}$, and $\mathcal{B}_n(\cdot)$ is the OBB regression head at stage n.

RRoI Align is designed to extract rotation-invariant features from arbitrarily oriented boxes. Given an oriented proposal parameterized by $(x_r, y_r, w_r, h_r, \theta_r)$, the region is divided into a regular grid of $K \times K$ cells. Within each cell (i, j), $m \times m$ sampling points are generated at normalized local coordinates

$$x = i\frac{w_r}{K} + (s_x + 0.5)\frac{w_r}{K \cdot m}, \quad y = j\frac{h_r}{K} + (s_y + 0.5)\frac{h_r}{K \cdot m} \tag{7}$$

for $s_x, s_y = 0, \ldots, m-1$. Each point (x, y) is then transformed into the global coordinate frame via an affine transformation that combines rotation and translation

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \mathcal{R}(\theta_r) \begin{pmatrix} x - \frac{w_r}{2} \\ y - \frac{h_r}{2} \end{pmatrix} + \begin{pmatrix} x_r \\ y_r \end{pmatrix} \tag{8}$$

where the rotation matrix $\mathcal{R}(\theta_r)$ is defined as follows:

$$\mathcal{R}(\theta_r) = \begin{pmatrix} \cos\theta_r & -\sin\theta_r \\ \sin\theta_r & \cos\theta_r \end{pmatrix}. \tag{9}$$

This transformation preserves the relative spatial structure under rotation. The features at the transformed coordinates (x', y') are sampled using bilinear interpolation from the feature map. The average of all m^2 samples within each cell forms the cell-level feature. Repeating this for all $K \times K$ cells produces a fixed-size feature map for each proposal. This process maintains spatial consistency even under significant rotation or small object scales, making it suitable for dense or highly oriented scenes.

To enhance the regression of OBBs and overcome the discontinuity introduced by angle periodicity, AFRNet encodes each OBB as a 2-D Gaussian distribution. An oriented box $(x_r, y_r, w_r, h_r, \theta_r)$ is mapped to a Gaussian $\mathcal{N}(\mu, \Sigma)$

$$\mu = \begin{pmatrix} x_r \\ y_r \end{pmatrix}, \quad \Sigma = \mathcal{R}(\theta_r) \begin{pmatrix} \frac{w_r^2}{4} & 0 \\ 0 & \frac{h_r^2}{4} \end{pmatrix} \mathcal{R}(\theta_r)^T \tag{10}$$

where μ denotes the box center and Σ encodes both shape and orientation. The standard deviations $w_r/2$ and $h_r/2$ correspond to the half-length and half-width of the rectangle, so that the unit Mahalanobis contour defined by $(x-\mu)^T\Sigma^{-1}(x-\mu) = 1$ forms an ellipse with axes (w_r, h_r), exactly matching the

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OBB [18]. This covariance-based encoding eliminates angular periodicity, preserves geometric consistency, and allows the box area to be recovered directly as $V(\Sigma) = 4|\Sigma|^{1/2} = w_r h_r$, thus ensuring exact correspondence to the rectangle.

During training, AFRNet uses a distribution-based regression loss, specifically the Kalman filter IoU (KFIoU) loss [47], to measure the similarity between predicted and ground-truth Gaussians. Unlike traditional regression losses that treat angle, scale, and position separately, this unified formulation jointly accounts for all geometric properties, resulting in more stable and robust learning. Empirically, this Gaussian modeling significantly improves orientation accuracy and overall detection performance, particularly for elongated or rotated targets.

3) Orientation-Aware Mask Head: To extend the detection capability of AFRNet from oriented object localization to instance-level segmentation, we append a dedicated mask prediction branch after the Cascade Gaussian-based RCNN. This branch is designed to generate high-resolution binary masks for each detected object, capturing fine-grained spatial boundaries while preserving orientation awareness.

For each refined oriented proposal, we apply RRoI Align to extract a 14×14 feature patch aligned with the orientation of the box. This alignment ensures that the spatial features are sampled consistently with the object's direction, which is particularly beneficial for nonaxis-aligned or densely packed objects. The extracted feature is then passed through four stacked 3×3 convolutional layers, followed by a 2×2 deconvolution layer with stride 2, resulting in an upsampled 28×28 feature map. A final 1×1 convolutional layer with K filters (where K is the number of object classes) produces K per-pixel mask logits—one for each class.

During training, only the mask corresponding to the ground-truth class k is supervised. The remaining $K-1$ class-specific masks are ignored. The mask loss is computed using a per-pixel binary cross-entropy loss between the predicted class k mask and the corresponding ground-truth binary mask.

At inference time, to reduce computational overhead, the mask branch is applied only to the top-scoring proposals. We then "undo" the RRoI Align: the 28×28 mask is resized and rotated back to the box's original size and orientation in the image, and finally thresholded at 0.5 to obtain a binary mask.

This orientation-aware mask head consistently produces high-quality and spatially accurate instance masks. By aligning the mask prediction with the object's orientation, it effectively suppresses background noise and enhances mask precision—particularly for rotated, elongated, or closely packed objects. Combined with the cascade's precise box localization, the mask branch enables AFRNet to deliver robust instance segmentation even in densely cluttered or complex scenes.

B. OBB-Guided SAM

The SAM supports promptable segmentation through input cues such as bounding boxes, points, or masks. However, its native design relies on HBBs, which may introduce substantial background or interobject interference when applied to rotated objects. An HBB enclosing a rotated object typically covers irrelevant background regions or adjacent objects. Moreover, directly using the OBBs as binary mask prompts can be counterproductive: SAM tends to treat the entire region as foreground, ignoring fine-grained object boundaries and including extraneous areas.

To address these challenges, we propose an OBB-guided SAM framework, which integrates high-quality OBBs from AFRNet without the orientation-aware mask head and employs a geometric prompting strategy tailored to SAM. As presented in Fig. 5, this approach effectively leverages orientation-aware localization from AFRNet and prompt-driven segmentation from SAM, enabling high-quality semisupervised instance masks for arbitrarily oriented objects.

Given an OBB with vertices $\{p_1, p_2, p_3, p_4\}$, we first compute the center point

$$p_+ = \frac{1}{4}\sum_{i=1}^{4} p_i \tag{11}$$

which serves as the foreground point prompt. To complement the foreground prompt, a set of negative point prompts is introduced to incorporate the surrounding spatial context of the OBB. Let $\mathbf{h}$ denote the minimal enclosing HBB. By combining the OBB with its enclosing HBB, four triangular regions are naturally formed. These regions typically correspond either to background clutter or to adjacent, densely arranged objects. Selecting the centroids of these triangles as negative points therefore provides an effective strategy to suppress interference. To compute these centroids, the vertices of both the OBB and HBB are merged into a single clockwise-ordered sequence $\{v_0, v_1, \ldots, v_7\}$. Starting from an index s corresponding to one of the OBB corners, four triangles are constructed by grouping three consecutive vertices, and their centroids are calculated as follows:

$$p_-^k = \frac{1}{3}\sum_{j=0}^{2} v_{(s+2k+j) \bmod 8}, \quad k \in \{0,1,2,3\}. \tag{12}$$

Here, the parameter s specifies the index of the initial OBB vertex. Since the four OBB vertices are geometrically equivalent, the choice of s only influences the ordering of the generated centroids without altering their spatial distribution. Consequently, the method is insensitive to the specific selection of s while ensuring consistent generation of negative point prompts. Each triangle centroid p_-^k thus serves as a negative point prompt, strategically located near the OBB boundaries to facilitate robust separation of the target foreground from surrounding background regions.

The complete SAM prompt consists of the following components.

1) *Box Prompt:* The minimal enclosing HBB $\mathbf{h}$.
2) *Point Prompt Set:* $\mathbf{p} = \{p_+, p_-^0, p_-^1, p_-^2, p_-^3\}$.
3) *Label Vector:* $\mathbf{l} = [1, 0, 0, 0, 0]$, indicating the first point as foreground and the rest as background.

Given an image $\mathbf{I}$, SAM produces an initial instance mask using these prompts

$$\mathbf{m}_{\text{sam}} = \text{SAM}(\mathbf{I}; \mathbf{h}, \mathbf{p}, \mathbf{l}). \tag{13}$$

To ensure that the output mask is consistent with the rotated geometry predicted by AFRNet, we apply a postprocessing

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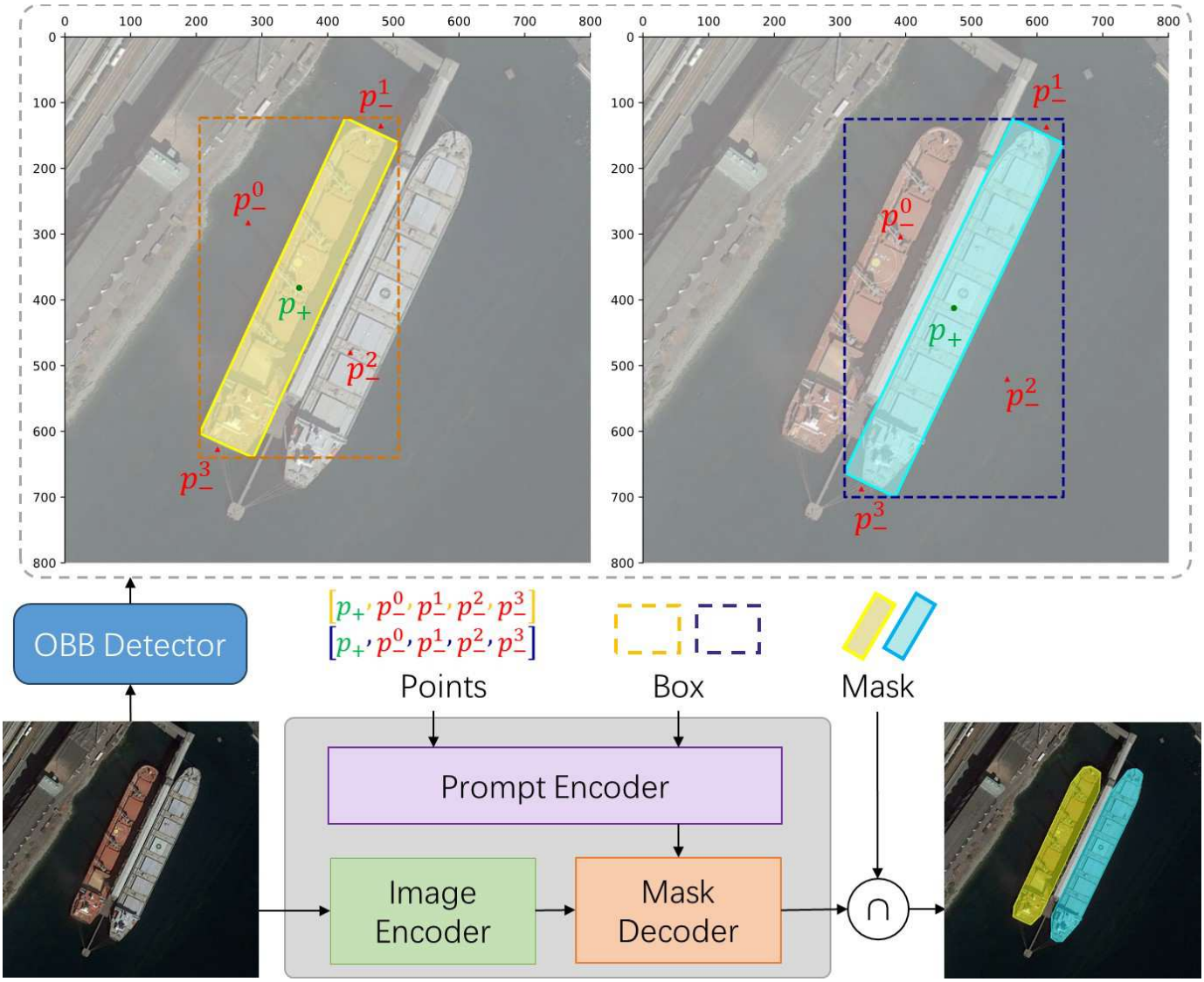


Fig. 5. Illustration of OBB-guided SAM. The process begins with the OBBs output from the detector, followed by generating four triangles from the circumscribed rectangles of the OBBs. The centroids of these triangles, along with the centroids of the OBBs, are calculated as prompts for SAM. The final results are produced by intersecting SAM's output with the OBBs.

refinement step. Specifically, we intersect the SAM output with the binary mask derived from the rotated box $\mathbf{m}_{\text{box}}$, yielding

$$\mathbf{m}_{\text{final}} = \mathbf{m}_{\text{sam}} \cap \mathbf{m}_{\text{box}}. \tag{14}$$

The proposed OBB-guided SAM approach integrates AFRNet's orientation-aware localization with SAM's powerful mask prediction capabilities. By combining axis-aligned and rotated box cues with carefully placed positive and negative points, the method mitigates common failure cases in prompt-based segmentations caused by background inclusion, mask leakage, and shape distortion. This framework enables high-quality semisupervised instance segmentation for rotated or elongated objects, even in cluttered scenes where traditional horizontal prompts fall short.

IV. Experiments

A. Datasets

To comprehensively evaluate the performance of our proposed method, we conduct experiments on six widely used benchmark datasets: HRSC2016 [48], SCCOS [49], DOTA [50], and DIOR-R [51] for rotated object detection, and VHR10-COCO [52] and iSAID [53] for instance segmentation. In addition, we apply our OBB-guided SAM strategy across all six datasets to assess its effectiveness in semisupervised instance segmentation.

HRSC2016 [48] and SCCOS [49] are both designed for rotated ship detection in maritime environments. HRSC2016 comprises 1,070 images with 2976 annotated ship instances, characterized by large aspect ratios and arbitrary orientations. SCCOS presents a more complex scenario, containing 4639 images and 19 078 instances. Compared to HRSC2016, it features higher object density, greater scale variation, more frequent occlusions, and a broader diversity of imaging conditions, offering a challenging testbed for robust and rotation-aware detection methods.

DOTA [50] and DIOR-R [51] are large-scale, multiclass benchmarks designed for rotated object detection in aerial imagery. DOTA contains 2806 aerial images collected from

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various sensors and platforms, annotated with over 188 000 objects across 15 categories. These span three domains: transportation vehicles, including small vehicle (SV), large vehicle (LV), plane (PL), helicopter (HC), and ship (SH); sports fields, including basketball court (BC), tennis court (TC), baseball diamond (BD), ground track field (GTF), soccer ball field (SBF), and swimming pool (SP); and architectural facilities, including bridge (BR), harbor (HA), roundabout (RA), and storage tank (ST). The image resolutions vary from 800×800 to 4000×4000 pixels and are typically split into smaller patches for training. DIOR-R comprises 23 463 high-resolution images (each sized 800×800) and 192 518 annotated instances across 20 object categories. These include transportation facilities such as airplane (APL), ship (SH), and vehicle (VE); sports and recreational areas such as baseball field (BF), BC, golf field (GF), GTF, stadium (STA), and TC; transport infrastructure such as airport (APO), bridge (BR), expressway service area (ESA), expressway toll station (ETS), harbor (HA), overpass (OP), and train station (TS); and industrial or utility structures such as chimney (CH), dam (DAM), storage tank (STO), and windmill (WM). Both datasets pose significant challenges due to wide variations in object orientation, scale, aspect ratio, and densely packed layouts, making them well-suited for evaluating the robustness of rotation-aware detection models.

VHR10-COCO [54] and iSAID [53] are two widely used benchmark datasets for instance segmentation in aerial imagery. VHR10-COCO provides reannotated mask labels for 650 optical remote sensing images from VHR10 [52], covering ten object categories: plane (PL), ship (SH), vehicle (VE), BD, BC, GTF, TC, bridge (BR), harbor (HA), and storage tank (ST). iSAID, a reannotated version of the DOTA dataset, contains 2806 high-resolution images with over 655 000 instance-level annotations across the same 15 categories as DOTA. Each object in iSAID is labeled with a precise, pixel-accurate mask. Compared to VHR10-COCO, iSAID presents substantially greater challenges due to arbitrarily oriented objects, large variations in scale, and dense spatial arrangements.

B. Evaluation

We adopt the COCO-style AP metric [55] to evaluate both detection and instance segmentation performance. Specifically, AP is computed across ten RIoU or IoU thresholds ranging from 0.50 to 0.95 in increments of 0.05. The average over these thresholds is denoted as $AP_{50:95}$

$$AP_{50:95} = \frac{1}{10} \sum_{t=0.50}^{0.95} AP_t. \tag{15}$$

We also report AP_{50} and AP_{75} to reflect performance under relaxed and stricter localization requirements, respectively.

For rotated object detection, we adopt the RIoU, defined as follows:

$$\mathrm{RIoU}(b_1, b_2) = \frac{\mathrm{Area}(b_1 \cap b_2)}{\mathrm{Area}(b_1 \cup b_2)} \tag{16}$$

where b_1 and b_2 denote two OBBs. The intersection area is computed by forming the polygon of overlap, triangulating it around a reference point, and summing the areas of the resulting triangles.

For instance segmentation, the IoU is calculated between predicted and ground-truth binary masks

$$\mathrm{IoU} = \frac{\left|m_{\mathrm{pred}} \cap m_{\mathrm{gt}}\right|}{\left|m_{\mathrm{pred}} \cup m_{\mathrm{gt}}\right|} \tag{17}$$

where m_{pred} and m_{gt} denote the predicted and ground-truth masks, respectively, and $|\cdot|$ represents the number of foreground pixels.

A prediction is considered correct if its RIoU or IoU exceeds the evaluation threshold. Given the numbers of true positives (TP), false positives (FP), and false negatives (FN), we compute precision and recall as follows:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \quad R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{18}$$

Varying confidence thresholds yields a precision–recall curve, and the area under this curve defines AP

$$\mathrm{AP}_t = \int_0^1 P_t(R)\, \mathrm{dR}. \tag{19}$$

This evaluation strategy enables consistent and rigorous benchmarking across rotated object detection (HRSC2016, SCCOS, DOTA, and DIOR-R) and instance segmentation (VHR10-COCO and iSAID).

In addition to detection accuracy, speed is a crucial factor in assessing a model's efficiency. A commonly used metric for evaluating detection speed is frames per second (FPS), which indicates the number of images processed per second.

C. Experiments Settings

Experiments are conducted on a single NVIDIA RTX 3090 GPU with a batch size of 2 for both training and inference, except for Mask2Former, BoxInst, DiscoBox, and Box2Mask, which are trained on an NVIDIA A800 GPU due to their high memory requirements. All models are implemented using the MMRotate [20] codebase. Following standard practice, we adopt horizontal and vertical flipping as data augmentation during training.

We adopt the stochastic gradient descent (SGD) optimizer with a momentum of 0.9, a weight decay of 0.0001, and an initial learning rate of 0.005. The training schedule is configured as follows: 12 epochs for DOTA and iSAID datasets, 36 epochs for SCCOS, DIOR-R, and VHR10-COCO, and 72 epochs for HRSC2016 dataset.

For input image resizing and preparation, different datasets are handled accordingly. In HRSC2016, images are resized such that the shorter side is 800 pixels while the longer side does not exceed 1333 pixels, and the aspect ratio is preserved. SCCOS images are kept at their original resolution of 1024×1024 pixels. In the case of DOTA, the original large-scale images are cropped into overlapping patches of size 1024×1024 pixels with a stride of 824 pixels, resulting in a 200-pixel overlap between adjacent patches. An NMS threshold of 0.1 is used during patch-level merging. For DIOR-R, images retain their original size of 800×800 pixels. In VHR10-COCO, images are resized such that the longer side

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TABLE I

COMPARISON WITH STATE-OF-THE-ART ROTATED OBJECT DETECTION METHODS ON HRSC2016, SCCOS, DOTA, AND DIOR-R. BOLD VALUES INDICATE THE BEST PERFORMANCE, AND UNDERLINED VALUES INDICATE THE SECOND BEST

Method	HRSC			SCCOS			DOTA			DIOR-R		
	AP_{50}	AP_{75}	$AP_{50:95}$	AP_{50}	AP_{75}	$AP_{50:95}$	AP_{50}	AP_{75}	$AP_{50:95}$	AP_{50}	AP_{75}	$AP_{50:95}$
Rotated Retinanet [25]	80.98	52.91	48.10	77.75	48.86	44.63	68.43	42.04	40.13	60.92	40.14	38.76
R3Det [26]	93.96	66.59	57.56	83.39	56.28	50.01	70.07	35.65	37.07	63.86	36.98	36.79
S2ANet [27]	92.39	71.24	59.59	86.89	65.64	56.11	73.90	35.53	39.05	66.18	36.08	37.16
DCFL [28]	95.73	75.99	63.75	86.32	64.86	56.70	74.91	45.54	44.07	68.27	40.44	40.22
Rotated FCOS [29]	91.43	72.37	60.92	84.62	63.30	55.16	71.27	37.08	39.42	65.09	41.04	40.03
PSC [56]	93.19	80.85	65.63	86.25	66.26	56.95	72.96	42.74	42.45	65.00	38.46	38.08
Rotated Faster R-CNN [4]	88.02	49.08	48.18	85.22	58.66	51.69	73.39	39.60	40.75	64.95	40.56	39.14
Gliding Vertex [23]	91.65	59.20	54.45	86.50	58.28	52.69	73.22	37.46	39.52	64.57	39.23	37.91
RoI Transformer [22]	94.70	81.66	67.95	90.17	76.46	63.89	76.13	46.88	45.58	67.33	44.48	42.43
QPDet [24]	96.56	85.04	68.78	88.84	73.98	61.26	74.59	45.59	43.89	65.71	42.19	40.36
Oriented R-CNN [15]	96.70	88.31	72.41	91.78	78.30	65.19	75.88	50.53	47.16	67.37	45.17	42.52
Ours	**98.36**	**94.09**	**80.34**	**91.88**	**82.36**	**69.13**	**78.87**	**55.41**	**50.69**	**72.01**	**52.27**	**48.92**

is 800 pixels, while preserving the original aspect ratio. For iSAID dataset, each image is divided into fixed-size patches of 800×800 pixels with a stride of 200 pixels.

Regarding dataset splits, we follow the official training and test partitions for all datasets except VHR10-COCO, which does not provide an official split. For this dataset, we randomly allocate 70% of the images for training and use the remaining 30% for testing. For DOTA and iSAID, since ground-truth annotations for the test set are not publicly available, we evaluate model performance through the official online evaluation servers.

D. Rotated Object Detection

Table I presents a comprehensive performance comparison between our proposed method (AFRNet without the orientation-aware mask head) and several state-of-the-art oriented object detection approaches on four benchmark datasets: HRSC2016, SCCOS, DOTA, and DIOR-R. For each method, we report three widely adopted evaluation metrics—AP_{50}, AP_{75}, and $AP_{50:95}$—to assess performance under varying localization thresholds.

On the HRSC2016 dataset, which focuses on ship detection with substantial orientation variations and extreme aspect ratios, our method achieves an outstanding AP_{50} of 98.36%, AP_{75} of 94.09%, and a strong $AP_{50:95}$ of 80.34%. Compared to the second-best approach, Oriented R-CNN (96.70%/88.31%/72.41%), our method delivers consistent improvements across all metrics, particularly under higher IoU thresholds, indicating enhanced precision and robustness in challenging geometric settings.

On the SCCOS dataset, characterized by even higher object density, greater scale variation, and more frequent occlusions, our method maintains state-of-the-art performance with 91.88% in AP_{50}, 82.36% in AP_{75}, and 69.13% in $AP_{50:95}$. While the gain in AP_{50} over Oriented R-CNN (91.78%) is modest, the improvements at stricter thresholds— +4.06% in AP_{75} and +3.94% in $AP_{50:95}$—underscore the effectiveness of our approach in accurately localizing rotated objects in cluttered maritime environments.

For the DOTA dataset, which is characterized by complex imaging conditions—including diverse sensor sources, varied scene contexts, and extreme variations in object orientation and aspect ratio—our method achieves 78.87% in AP_{50}, 55.41% in AP_{75}, and 50.69% in $AP_{50:95}$. Although several detectors, such as RoI Transformer, QPDet, and Oriented R-CNN, perform competitively under relaxed IoU criteria, their performance degrades significantly at stricter thresholds. In contrast, our model maintains robust accuracy across all IoU levels, demonstrating superior localization precision and robustness under challenging, real-world remote sensing conditions.

On the more diverse DIOR-R dataset, which comprises 20 object classes with substantial variability in scale, density, and semantic context, our approach again establishes a new state of the art. It achieves 72.01% in AP_{50}, 52.27% in AP_{75}, and 48.92% in $AP_{50:95}$, yielding absolute improvements of +4.64%, +7.10%, and +6.40% over the second-best model, Oriented R-CNN. Table II further reports detailed category-wise $AP_{50:95}$ along with FPS. As shown, improvements are consistent across nearly all object categories. In particular, our approach achieves substantial gains in challenging categories such as airplane (+10.67%), ETS (+8.04%), ESA (+9.02%), and TS (+7.52%). This demonstrates strong robustness to category-specific variations in shape, scale, and orientation, while incurring only a minor FPS drop of 2.6 compared to Oriented R-CNN, and highlights the generalization capability of our method across heterogeneous object types and diverse remote sensing scenes.

Overall, the results clearly demonstrate that our method consistently outperforms prior detectors across all datasets and evaluation thresholds. The most significant gains are observed in the stricter IoU metrics (AP_{75} and $AP_{50:95}$), which are particularly sensitive to bounding box alignment quality. This confirms the strength of our proposed rotation-aware architecture in accurately capturing object geometry, and its robustness in handling dense, complex, and large-scale aerial imagery. These consistent improvements across diverse benchmarks validate the broad applicability and generalization ability of our method for rotated object detection in remote sensing scenarios.

E. Instance Segmentation

Tables III and IV present comparative results of our proposed instance segmentation framework (AFRNet with the

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TABLE II

COMPARISON WITH STATE-OF-THE-ART ROTATED OBJECT DETECTION METHODS ON DIOR-R, REPORTED WITH CLASSWISE AP. BOLD VALUES INDICATE THE BEST PERFORMANCE, AND UNDERLINED VALUES INDICATE THE SECOND BEST

Methods	APL	APO	BF	BC	BR	CH	DAM	ETS	ESA	GF	GTF	HA	OP	SH	STA	STO	TC	TS	VE	WM	$AP_{50:95}$	FPS
Rotated Retinanet [25]	38.43	8.94	56.50	74.01	12.23	62.88	6.78	43.97	42.05	45.28	62.25	18.07	23.77	49.83	46.99	37.48	75.45	21.79	21.96	26.51	38.76	**44.7**
R3Det [26]	36.69	11.68	52.80	65.08	15.63	59.27	6.94	42.89	41.12	38.58	57.45	17.10	27.98	46.65	45.09	37.27	67.97	20.22	21.81	23.70	36.79	31.5
S2ANet [27]	34.36	13.24	42.32	67.47	17.98	53.23	7.29	44.87	39.57	37.59	59.59	19.54	29.95	53.18	42.10	36.06	71.51	22.25	25.81	25.26	37.16	37.6
DCFL [28]	39.56	16.07	56.73	69.40	18.03	60.90	9.08	46.70	41.25	44.37	60.26	21.22	31.01	53.35	42.36	41.59	73.63	25.03	25.45	28.33	40.22	40.8
Rotated FCOS [29]	39.77	12.82	55.82	73.44	18.11	62.69	6.32	46.01	41.65	43.38	59.96	19.57	30.73	54.69	42.98	41.80	74.47	23.13	24.96	28.39	40.03	44.3
PSC [56]	37.59	13.10	54.33	66.75	17.32	61.84	6.07	46.86	42.09	40.68	55.47	17.65	29.51	50.69	42.14	40.05	66.70	20.39	23.96	28.49	38.08	41.0
Rotated Faster R-CNN [4]	38.48	9.21	54.26	70.29	14.89	60.07	5.95	42.83	36.98	42.58	61.43	22.38	27.30	54.96	52.01	40.76	71.23	25.18	23.86	28.19	39.14	39.3
Gliding Vertex [23]	36.86	8.39	53.71	66.40	14.29	60.02	5.16	42.69	36.26	42.28	58.15	21.32	25.88	52.53	48.43	39.86	68.03	25.25	23.95	28.71	37.91	38.0
RoI Transformer [22]	42.51	14.74	53.93	73.99	19.69	60.85	9.25	48.38	45.38	42.84	65.67	25.49	33.14	57.56	56.70	41.81	74.98	25.27	26.94	29.54	42.43	32.0
QPDet [24]	38.62	12.76	54.96	69.58	19.30	60.09	9.84	46.06	43.57	42.00	61.72	21.87	31.00	55.09	49.11	**42.71**	71.33	24.37	25.27	27.98	40.36	32.7
Oriented R-CNN [15]	42.69	15.12	56.80	73.14	21.63	61.31	9.99	49.08	47.25	42.74	64.94	24.49	34.36	58.21	51.10	40.77	74.88	27.23	26.51	28.22	42.52	32.9
Ours	**53.36**	**22.40**	**63.47**	**81.37**	**27.96**	**66.10**	**14.41**	**57.12**	**56.27**	**49.27**	**72.15**	**33.35**	**40.56**	**62.49**	**57.80**	42.39	**80.19**	**34.75**	**30.78**	**32.24**	**48.92**	30.3

TABLE III

COMPARISON WITH STATE-OF-THE-ART INSTANCE SEGMENTATION METHODS ON VHR10-COCO. BOLD VALUES INDICATE THE BEST PERFORMANCE, AND UNDERLINED VALUES INDICATE THE SECOND BEST

Method	PL	BD	BR	GTF	VE	SH	TC	BC	ST	HA	AP_{50}	AP_{75}	$AP_{50:95}$
YOLACT [12]	18.63	78.06	26.68	77.16	48.18	52.86	30.86	53.99	77.21	28.85	84.08	50.96	49.25
SOLOv2 [32]	23.23	69.62	28.96	86.59	55.88	52.97	54.28	66.42	42.62	22.71	82.39	51.68	50.33
Mask2Former [14]	36.12	80.26	41.93	94.23	59.89	54.93	70.74	71.37	45.95	45.68	84.20	65.71	60.11
Mask R-CNN [5]	39.57	84.56	40.13	91.29	64.99	61.58	76.81	74.59	83.06	61.35	93.90	75.12	67.79
HRNet [6]	43.95	**87.36**	41.41	91.20	66.47	61.22	78.78	74.90	83.33	60.39	95.05	75.48	68.90
GROI [8]	38.03	85.75	42.15	92.66	64.75	58.29	76.55	75.04	83.80	63.82	94.37	73.49	68.08
Mask Scoring R-CNN [9]	42.11	84.84	42.97	92.57	64.80	62.49	78.33	76.31	81.20	63.97	94.03	76.87	68.96
Cascade Mask R-CNN [10]	39.59	86.08	43.73	93.10	64.15	61.71	77.87	76.44	82.79	62.97	94.23	76.71	68.84
HTC [11]	41.77	86.22	43.45	92.80	65.72	64.27	78.88	78.01	82.19	64.85	95.71	76.73	69.82
SCNet [31]	44.71	86.39	38.16	94.24	63.64	63.91	77.27	77.72	81.76	65.86	94.10	77.09	69.37
Oriented Mask R-CNN [15]	44.95	86.52	43.03	91.29	65.24	67.22	80.02	78.04	82.40	66.13	94.62	79.74	70.48
Ours	**49.94**	87.28	**50.94**	**94.82**	**69.20**	**68.84**	**84.07**	**80.44**	**84.29**	**68.44**	**96.89**	**83.37**	**73.83**

TABLE IV

COMPARISON WITH STATE-OF-THE-ART INSTANCE SEGMENTATION METHODS ON iSAID. BOLD VALUES INDICATE THE BEST PERFORMANCE, AND UNDERLINED VALUES INDICATE THE SECOND BEST

Method	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	AP_{50}	AP_{75}	$AP_{50:95}$	FPS
YOLACT [12]	26.31	33.71	8.89	10.54	10.24	15.96	30.32	53.54	28.08	25.02	9.02	38.05	17.23	29.82	4.73	42.86	21.60	22.76	30.8
SOLOv2 [32]	36.84	42.15	12.66	18.70	7.35	19.17	29.11	69.02	33.93	24.53	18.23	26.80	26.49	30.81	10.40	48.07	26.65	27.08	25.7
Mask2Former [14]	39.44	44.05	12.57	26.04	9.90	23.03	33.52	71.55	35.91	26.34	31.03	29.82	29.19	33.91	11.35	50.10	31.84	30.51	16.1
Mask R-CNN [5]	44.57	50.68	18.02	27.98	14.68	30.55	47.29	76.69	51.86	35.95	30.53	43.23	31.21	38.30	9.41	58.91	39.68	36.73	**31.7**
HRNet [6]	46.08	55.83	20.67	33.46	14.89	30.51	47.49	77.38	54.34	36.68	38.82	45.98	32.98	38.72	8.50	61.67	41.83	38.82	21.8
GROI [8]	45.22	56.24	20.02	30.93	14.53	31.76	47.92	77.00	53.36	34.61	36.31	43.73	32.55	36.94	10.28	61.39	41.05	38.09	16.7
Mask Scoring R-CNN [9]	45.25	57.62	19.13	30.80	14.39	31.34	48.03	77.29	54.66	35.65	35.34	47.31	33.05	38.64	12.56	60.52	42.22	38.73	29.9
Cascade Mask R-CNN [10]	46.64	55.22	18.31	31.27	14.92	32.28	49.05	77.20	53.36	35.60	38.68	45.84	33.84	39.99	12.35	60.41	42.84	38.97	23.8
HTC [11]	45.99	57.93	20.72	32.29	14.69	32.77	48.70	78.56	54.61	36.61	40.42	46.52	33.61	39.03	10.52	62.68	42.59	39.53	18.3
SCNet [31]	46.56	55.58	17.90	28.29	14.64	32.56	47.89	78.32	51.16	36.17	33.83	**47.55**	32.12	39.68	**14.25**	60.67	41.56	38.43	17.6
Oriented Mask R-CNN [15]	43.36	57.26	20.21	31.76	15.44	33.39	49.16	76.74	51.75	36.21	36.29	44.58	32.03	38.16	12.68	63.43	41.38	39.05	23.6
Ours	**47.67**	**58.56**	**20.97**	**36.51**	**16.48**	**35.36**	**51.57**	**78.87**	**57.60**	**37.97**	**41.55**	45.77	**35.80**	**41.35**	13.18	**65.90**	**44.95**	**41.76**	20.2

orientation-aware mask head) against a variety of state-of-the-art methods on the VHR10-COCO and iSAID datasets, respectively.

On the VHR10-COCO dataset (Table III), our method achieves the best performance across all evaluation metrics, obtaining AP_{50} of 96.89%, AP_{75} of 83.37%, and $AP_{50:95}$ of 73.83%. Compared to the second-highest values in each metric—95.71% for AP_{50} (HTC), 79.74% for AP_{75} (Oriented Mask R-CNN), and 70.48% for $AP_{50:95}$ (Oriented Mask R-CNN)—our approach delivers absolute gains of +1.18%, +3.63%, and +3.35%, respectively. These gains are especially notable under stricter IoU thresholds, demonstrating enhanced localization precision and robustness. At the category level, substantial performance gains are observed in challenging classes such as plane (+4.99%), bridge (+7.21%), and TC (+4.05%), where object orientation varies significantly. These results validate the effectiveness of our rotation-aware design in capturing fine-grained, alignment-consistent masks. Representative segmentation results are visualized in Fig. 6. Note that, in VHR10-COCO, the harbor category encompasses both harbor areas and the ships within them.

On the iSAID dataset (Table IV), which presents substantial challenges due to densely packed objects and diverse object categories, our method achieves the best performance across all evaluation metrics. It attains an AP_{50} of 65.90%, AP_{75} of 44.95%, and an $AP_{50:95}$ of 41.76%, yielding absolute

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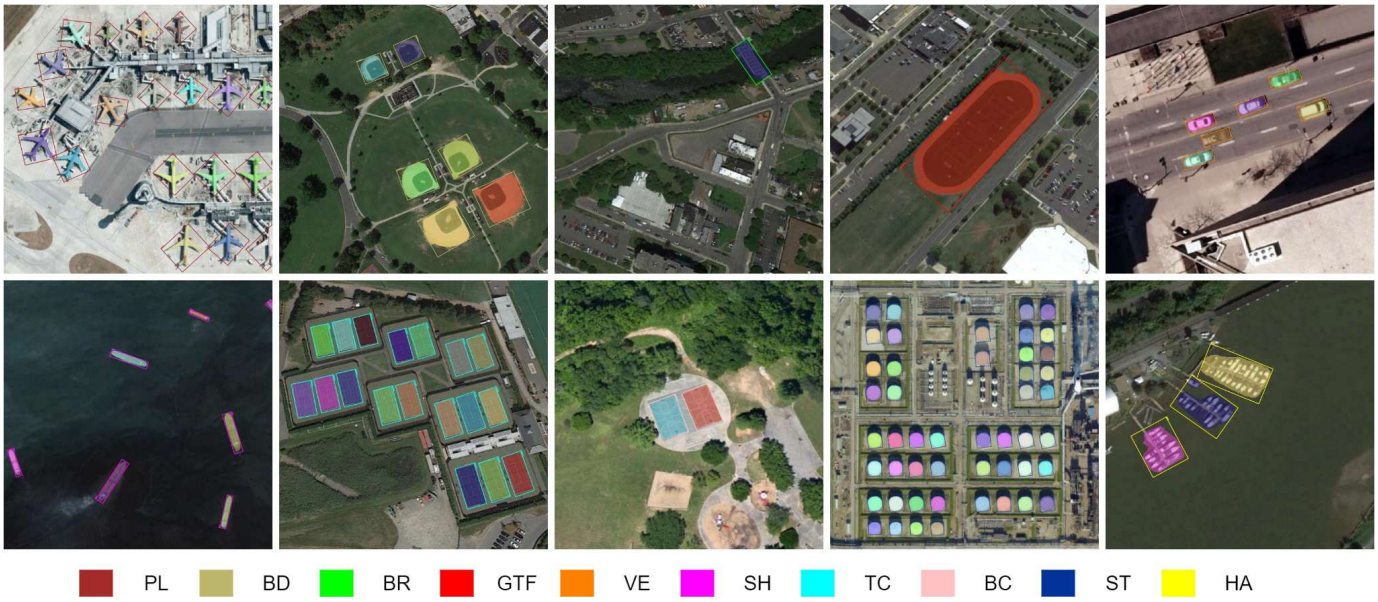


Fig. 6. Example instance segmentation results of AFRNet on VHR10-COCO. OBBs are drawn in distinct colors to represent different object classes, with the legend below specifying the mapping between OBB colors and categories.

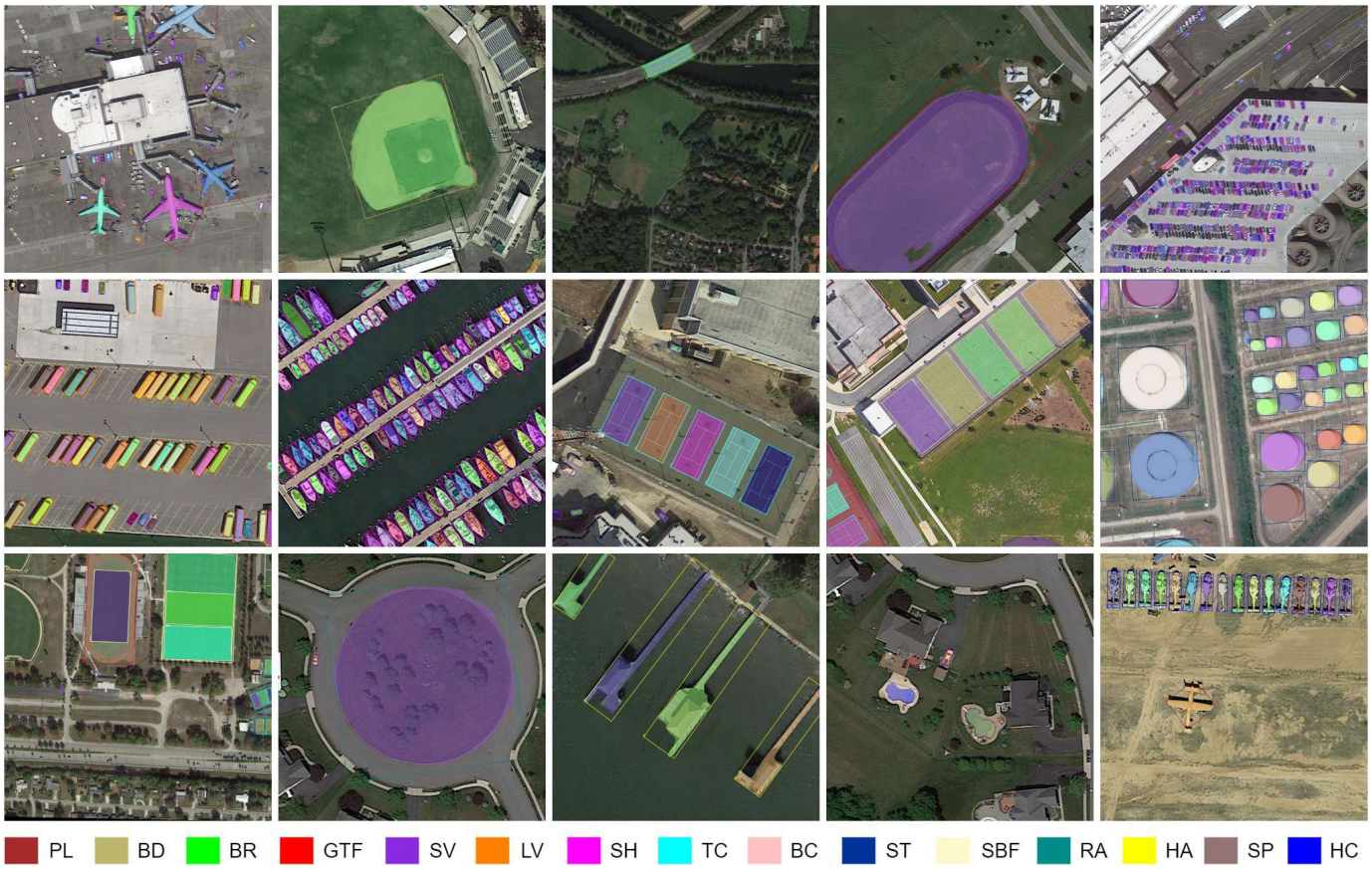


Fig. 7. Example instance segmentation results of AFRNet on iSAID. OBBs are drawn in distinct colors to represent different object classes, with the legend below specifying the mapping between OBB colors and categories.

improvements of +2.47%, +2.11%, and +2.23% over the second-best values—63.43% for AP_{50} (Oriented Mask R-CNN), 42.84% for AP_{75} (Cascade Mask R-CNN), and 39.53% for $AP_{50:95}$ (HTC). Notably, compared to the AP_{50} second-best method (Oriented Mask R-CNN), our FPS drops by only 3.4, while compared to the $AP_{50:95}$ second-best method (HTC), it increases by 1.9, demonstrating a favorable balance between accuracy and efficiency. In addition to overall performance,

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TABLE V

COMPARISON OF SEMISUPERVISED INSTANCE SEGMENTATION ON VHR10-COCO. BOLD VALUES INDICATE THE BEST PERFORMANCE, AND UNDERLINED VALUES INDICATE THE SECOND BEST

Method	PL	BD	BR	GTF	VE	SH	TC	BC	ST	HA	AP_{50}	AP_{75}	$AP_{50:95}$
Mask R-CNN [5]	**39.57**	**84.56**	**40.13**	91.29	**64.99**	61.58	**76.81**	**74.59**	**83.06**	**61.35**	**93.90**	**75.12**	**67.79**
BoxInst [37]	7.18	71.94	13.26	88.61	42.88	45.65	57.00	50.86	68.23	19.85	80.18	44.87	46.55
DiscoBox [38]	12.12	72.40	7.21	82.88	39.49	40.93	56.14	49.22	36.12	17.61	79.61	41.13	41.41
Box2Mask [39]	12.04	76.93	22.55	88.74	49.23	42.43	64.89	66.37	31.94	41.60	80.31	51.87	49.72
HBB-Guided SAM [19]	8.83	75.16	8.05	74.38	58.04	44.48	69.18	69.18	67.43	54.00	81.34	60.09	52.87
Ours	14.64	83.22	9.29	**92.67**	61.73	**62.92**	64.96	71.78	76.07	52.46	88.38	67.16	58.97

TABLE VI

COMPARISON OF SEMISUPERVISED INSTANCE SEGMENTATION ON iSAID. BOLD VALUES INDICATE THE BEST PERFORMANCE, AND UNDERLINED VALUES INDICATE THE SECOND BEST

Method	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	AP_{50}	AP_{75}	$AP_{50:95}$
Mask R-CNN [5]	**44.57**	50.68	**18.02**	27.98	**14.68**	30.55	47.29	**76.69**	**51.86**	35.95	30.53	43.23	**31.21**	38.30	9.41	58.91	**39.68**	**36.73**
BoxInst [37]	16.99	27.53	3.81	10.90	6.76	17.66	25.64	37.55	23.48	23.68	16.11	21.60	14.15	26.22	5.85	43.96	12.85	18.53
DicoBox [38]	7.19	48.68	7.92	21.32	5.33	14.97	26.18	71.83	45.76	23.28	25.23	34.82	7.25	26.97	6.20	44.98	23.58	24.86
Box2Mask [39]	20.64	49.71	7.76	22.49	4.27	14.67	21.30	72.13	45.48	21.51	29.02	32.02	17.35	23.90	10.87	51.38	22.95	26.21
HBB-Guided SAM [19]	26.29	45.90	10.82	21.06	10.89	26.64	40.92	69.33	40.46	35.16	32.79	43.37	26.20	27.42	12.33	56.97	29.58	31.31
Ours	25.11	**54.22**	13.63	**30.16**	14.50	**31.70**	**48.61**	65.49	39.57	**37.15**	**38.27**	**44.52**	25.32	**38.41**	**12.39**	**61.74**	34.94	35.03

our method ranks first in 13 of 15 categories, with consistent gains in orientation-sensitive classes such as SV (+1.04%), LV (+1.97%), ship (+2.41%), and harbor (+1.96%). These categories are particularly sensitive to object orientation and tight layouts, and the consistent gains underscore the effectiveness of our rotation-aware feature refinement and mask prediction strategy. Representative segmentation results are visualized in Fig. 7.

Overall, the results on both datasets confirm that our model achieves state-of-the-art performance in instance segmentation under arbitrary orientations and challenging remote sensing conditions. Specifically, VHR10-COCO is a relatively small dataset, while iSAID presents a much larger and more complex benchmark with dense object distributions and diverse categories. Despite these differences, our model consistently achieves top performance across both scales, demonstrating strong generalization ability. The consistent gains across multiple categories and evaluation metrics further validate the effectiveness of our rotation-aware design, particularly under high-precision segmentation demands.

F. Semisupervised Instance Segmentation

Tables V and VI report the results of semisupervised instance segmentation on the VHR10-COCO and iSAID datasets, respectively. In this setting, the objective is to generate high-quality instance masks using only limited or weak annotations. We compare the standard fully supervised Mask R-CNN (serving as an upper bound), BoxInst, DiscoBox, Box2Mask, the HBB-guided SAM, and our proposed OBB-guided SAM framework.

On the VHR10-COCO dataset (Table V), our OBB-guided SAM consistently outperforms other semisupervised methods across nearly all evaluation metrics. In particular, it achieves an AP_{50} of 58.97%, compared with 52.87% for the second-best HBB-guided SAM, thereby narrowing the gap to the fully supervised Mask R-CNN baseline (67.79%). Under the relaxed AP_{50} metric, the gap becomes even smaller (88.38% versus 93.90%). Notably, in certain orientation-sensitive categories such as ship and GTF, our OBB-guided approach even surpasses the fully supervised counterpart. These results highlight the effectiveness of incorporating rotation-aware geometric cues into prompt-based segmentation.

A similar trend is observed on the more challenging iSAID dataset (Table VI), which contains densely packed instances across 15 diverse categories. Our OBB-guided SAM achieves an $AP_{50:95}$ of 35.03%, closely approaching that of the fully supervised Mask R-CNN (36.73%), while significantly outperforming the second-best HBB-guided SAM (31.31%). The gap to full supervision is only 1.7% points under $AP_{50:95}$, and our OBB-guided method even surpasses Mask R-CNN under AP_{50} (61.74% versus 58.91%). Furthermore, the gains are consistent across a wide range of categories, particularly in GTF, SV, LV, and ship, highlighting the robustness of our method in complex aerial scenes.

In summary, the proposed OBB-guided SAM framework offers a highly effective semisupervised alternative to fully supervised instance segmentation models. Beyond standard segmentation benchmarks, we further apply our framework to several rotated object detection datasets—HRSC2016, SCCOS, DOTA, and DIOR-R. Although these datasets lack ground-truth mask annotations for quantitative evaluation, we present qualitative visualizations of the instance masks produced through semisupervised learning, as shown in Figs. 8–10. These examples highlight the ability of our method to produce accurate, rotation-aligned masks in diverse remote sensing scenarios. By leveraging orientation-aware geometric prompts derived from detection outputs, our approach enables high-quality instance segmentation without requiring pixel-level supervision, making it well-suited for real-world applications with limited annotation resources.

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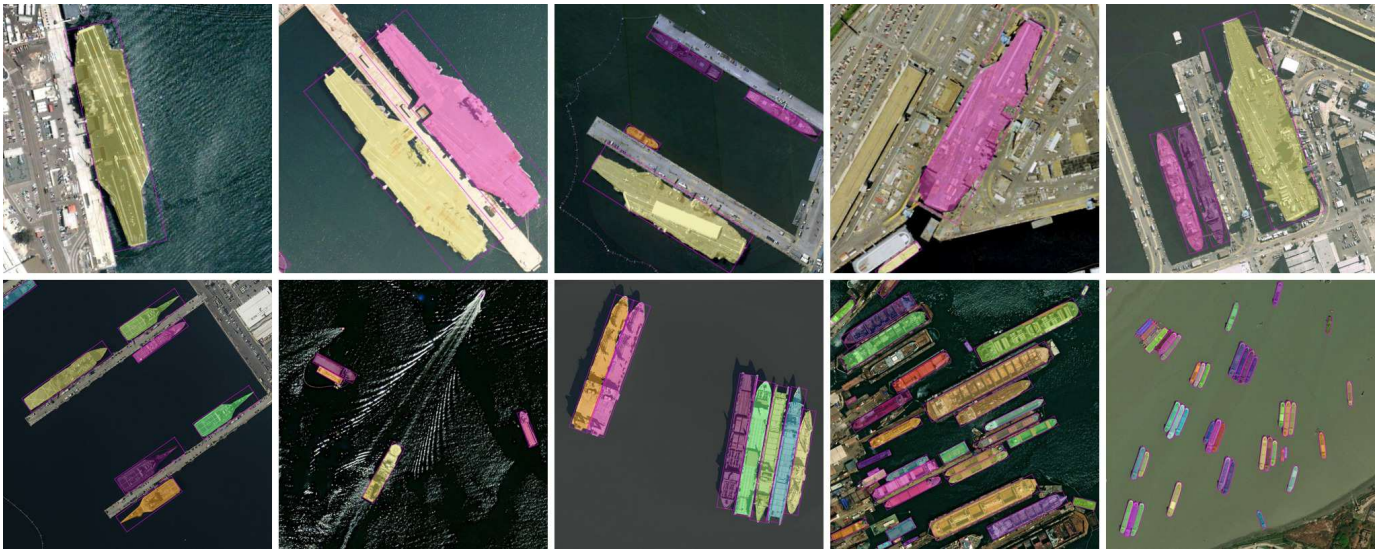


Fig. 8. Example semisupervised segmentation results of the OBB-guided SAM on HRSC2016 and SCCOS. Results from HRSC2016 are shown in the first row, and those from SCCOS appear in the second row.

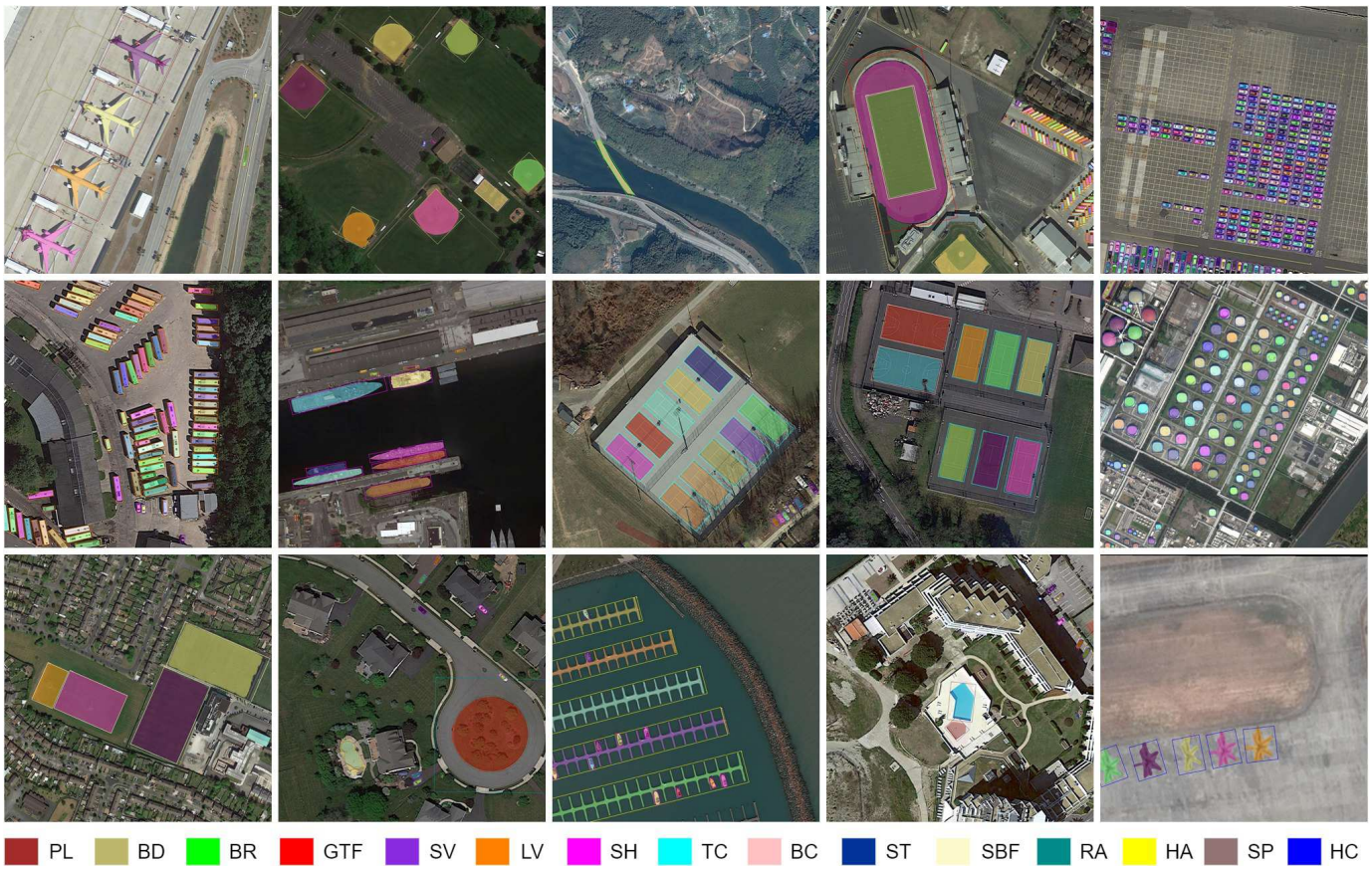


Fig. 9. Example semisupervised segmentation results of the OBB-guided SAM on DOTA. OBBs are drawn in distinct colors to represent different object classes, with the legend below specifying the mapping between OBB colors and categories.

G. Ablation Study

To evaluate the contributions of core components within the AFRNet framework, we conduct a series of ablation studies on DIOR-R for rotated object detection and iSAID for instance segmentation. Specifically, we examine three aspects: 1) an incremental ablation of AFRNet with different components (DARPN and the Gaussian-based loss); 2) the effect of using DCN versus modulated DCN within DARPN; and 3) the influence of the number of cascade refinement stages on both detection and segmentation performance, as well as inference efficiency.

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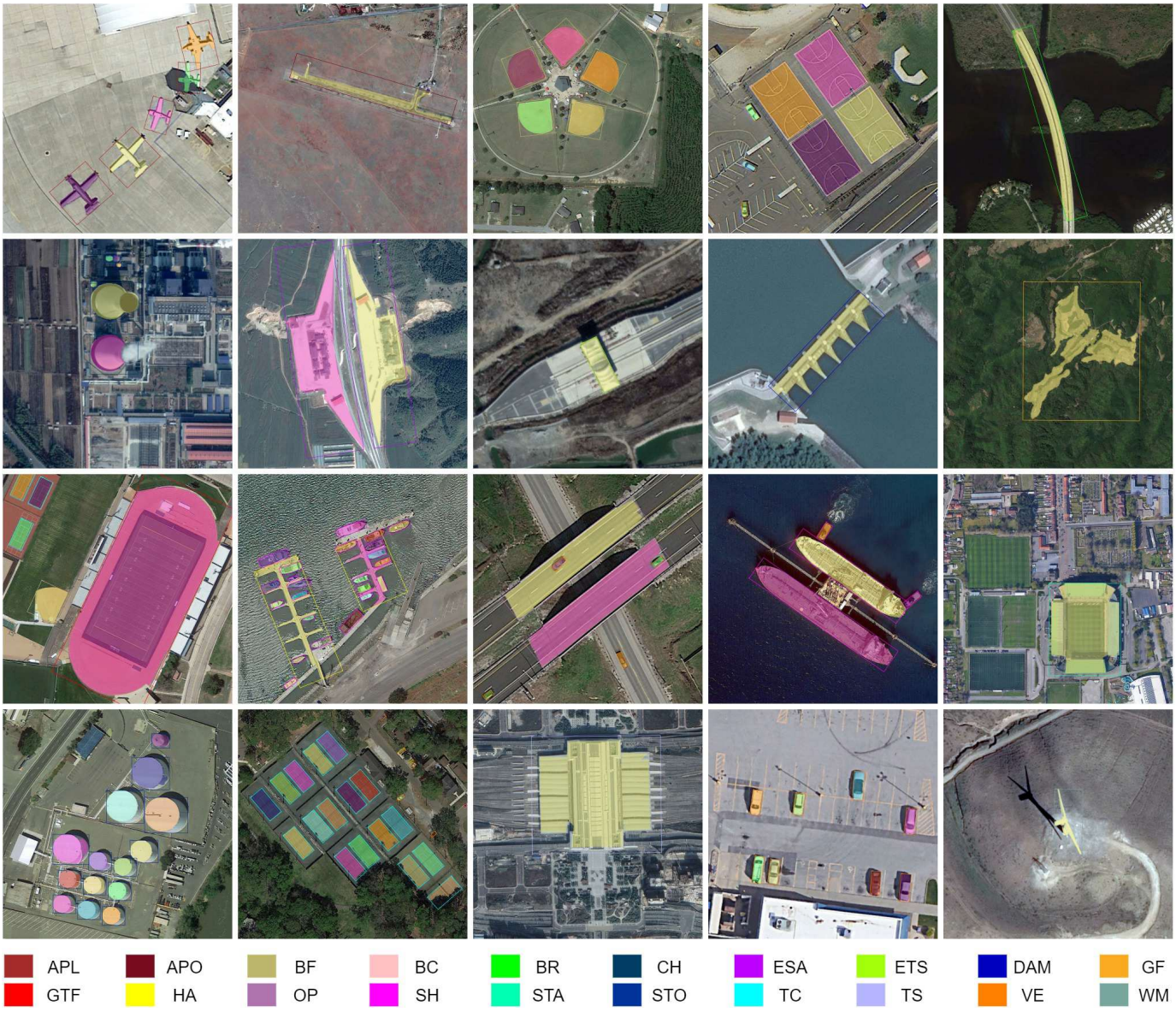


Fig. 10. Example semisupervised segmentation results of the OBB-guided SAM on DIOR-R. OBBs are drawn in distinct colors to represent different object classes, with the legend below specifying the mapping between OBB colors and categories.

TABLE VII
INCREMENTAL ABLATION STUDY ON DIOR-R AND iSAID. BOLD VALUES INDICATE THE BEST PERFORMANCES

Method	DIOR-R				iSAID			
	AP_{50}	AP_{75}	$AP_{50:95}$	FPS	AP_{50}	AP_{75}	$AP_{50:95}$	FPS
AFRNet	70.24	50.29	47.12	**31.2**	63.94	43.65	40.43	**20.8**
+ Gaussian-based Loss	71.86	51.07	47.96	31.1	65.11	44.22	41.13	20.8
+ DARPN	71.49	51.43	48.28	30.4	64.68	44.73	41.37	20.2
+ DARPN + Gaussian-based Loss	**72.01**	**52.27**	**48.92**	30.3	**65.90**	**44.95**	**41.76**	20.2

Table VII presents the results of the incremental ablation study. Starting from the baseline AFRNet with a single refinement stage (without DARPN or Gaussian-based loss), we observe that the addition of each component leads to consistent improvements on both the DIOR-R and iSAID datasets. Specifically, introducing the Gaussian-based loss alone increases AP_{50} from 70.24% to 71.86% and $AP_{50:95}$ from 47.12% to 47.96% on DIOR-R, and from 63.94% to 65.11% and 40.43% to 41.13% on iSAID. This indicates that modeling-oriented bounding box regression with a Gaussian-based loss allows the network to better capture object shapes and orientations, leading to more accurate localization with almost no loss in inference speed. Introducing DARPN with modulated DCN alone also yields consistent

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TABLE VIII
EFFECT OF DCN VERSUS MODULATED DCN IN DARPN ON DIOR-R AND iSAID. BOLD VALUES INDICATE THE BEST PERFORMANCES

Method	DIOR-R				iSAID			
	AP_{50}	AP_{75}	$AP_{50:95}$	FPS	AP_{50}	AP_{75}	$AP_{50:95}$	FPS
DCN	71.92	51.53	48.19	**30.3**	65.02	44.52	41.28	**20.2**
Modulated DCN	**72.01**	**52.27**	**48.92**	30.3	**65.90**	**44.95**	**41.76**	20.2

TABLE IX
IMPACT OF THE NUMBER OF CASCADE REFINEMENT STAGES ON DIOR-R AND iSAID. BOLD VALUES INDICATE THE BEST PERFORMANCES

# Stages	DIOR-R				iSAID			
	AP_{50}	AP_{75}	$AP_{50:95}$	FPS	AP_{50}	AP_{75}	$AP_{50:95}$	FPS
1	**72.01**	52.27	48.92	**30.3**	**65.90**	44.95	41.76	**20.2**
2	71.85	53.24	49.64	25.8	65.58	**45.61**	**42.22**	17.9
3	71.41	**53.28**	**49.84**	22.3	65.33	45.33	41.96	16.5

gains, raising $AP_{50:95}$ to 48.28% on DIOR-R and 41.37% on iSAID, which highlights the benefit of deformable proposal refinement for handling arbitrarily oriented objects. When both DARPN and the Gaussian-based loss are combined, the network achieves the highest performance, with $AP_{50:95}$ reaching 48.92% on DIOR-R and 41.76% on iSAID, while maintaining competitive inference speed (30.3 FPS on DIOR-R and 20.2 FPS on iSAID). Overall, these results demonstrate that each component contributes positively to both detection and segmentation performance, and their combination yields synergistic improvements.

Table VIII compares the performance of standard DCN and modulated DCN within the DARPN module on DIOR-R and iSAID under the configuration of a single refinement stage with the Gaussian-based loss. The results show that modulated DCN consistently outperforms standard DCN across all evaluation metrics, with minimal impact on inference speed. These findings indicate that the modulated offsets in modulated DCN more effectively capture geometric variations of arbitrarily oriented objects, thereby enhancing proposal refinement within DARPN without sacrificing efficiency.

In Table IX, we evaluate the effect of increasing the number of cascade refinement stages from one to three, with both DARPN and Gaussian-based loss integrated. In our cascade design, each stage increases the IoU threshold by 0.1, with the final-stage threshold set to 0.6 for one refinement stage, 0.7 for two, and 0.8 for three. Transitioning from one to two refinement stages leads to a modest but consistent performance improvement, particularly under stricter localization criteria. For example, AP_{75} increases from 52.27% to 53.24% on DIOR-R and from 44.95% to 45.61% on iSAID. Although a configuration with three refinement stages yields the highest $AP_{50:95}$ on DIOR-R (49.84%), the improvement over two stages is marginal (0.2% points) and comes at the cost of a drop in inference speed (from 25.8 to 22.3 FPS). Notably, the performance on iSAID slightly declines with three refinement stages. This drop can be attributed to the increasing IoU thresholds in deeper stages: in this configuration, the last stage uses a high IoU threshold of 0.8. While this improves predictions at high IoUs (e.g., AP_{80} and above), it can reduce AP at lower IoUs (e.g., AP_{50} and AP_{75}). Since $AP_{50:95}$ averages over multiple IoU thresholds, the gains at high IoUs do not fully compensate for the reductions at low IoUs, resulting in a slight decrease in overall AP. Note that iSAID results are from online evaluation and report only AP_{50}, AP_{75}, and $AP_{50:95}$. These observations suggest that blindly increasing the number of cascade stages does not always improve performance and may introduce diminishing returns or efficiency trade-offs.

H. Discussion

Our extensive experiments across six benchmark datasets reveal several important insights into the capabilities and limitations of our proposed AFRNet framework and its associated components for rotated object detection and instance segmentation in aerial imagery.

1) Effectiveness of Rotation-Aware Detection: The results in Table I demonstrate that AFRNet achieves state-of-the-art performance across four widely used rotated object detection benchmarks: HRSC2016, SCCOS, DOTA, and DIOR-R. On each dataset, AFRNet consistently outperforms prior methods across all AP metrics. These gains can be attributed to three key architectural innovations: 1) the use of DARPN for generating spatially adaptive and rotation-sensitive proposals; 2) a Gaussian-based representation of OBBs that embeds geometry into a continuous and differentiable space; and 3) a cascade refinement framework that progressively improves box quality under stricter RIoU thresholds. Ablation results in Table VII confirm the effectiveness of each component, showing steady performance gains from DARPN and Gaussian modeling. Table VIII further demonstrates that modulated DCN consistently outperforms standard DCN within the DARPN module, enhancing proposal refinement with minimal impact on inference speed. Additionally, Table IX reveals that increasing the number of cascade stages enhances detection accuracy, particularly under high IoU thresholds. Together, these findings highlight the efficiency and precision of our rotation-aware design.

2) Instance Segmentation Quality and Alignment: Tables III and IV report instance segmentation results on VHR10-COCO and iSAID, respectively. AFRNet outperforms all existing baselines in terms of AP_{50}, AP_{75}, and $AP_{50:95}$, confirming the effectiveness of the orientation-aware mask head. The use of RRoI Align ensures that the features used for mask prediction are spatially aligned with object geometry, resulting in more accurate and cleaner masks, particularly for small or overlapping objects. Notably, our model performs especially well on categories with strong rotation sensitivity,

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demonstrating the benefit of aligning mask prediction with object orientation.

3) Semisupervised Segmentation With OBB-Guided SAM: In the semisupervised setting, where only box-level annotations are available, our OBB-guided SAM framework provides a powerful alternative to traditional mask-supervised training. As shown in Tables V and VI, OBB-guided prompting consistently outperforms the HBB-guided baseline across all metrics and datasets. Notably, on the iSAID dataset, it achieves an $AP_{50:95}$ of 35.03%, reducing the gap to the fully supervised Mask R-CNN (36.73%) to less than 2% points. The performance gains are particularly evident in categories with nonaxis-aligned objects and densely packed layouts—scenarios where precise orientation cues are critical. These results highlight the importance of incorporating orientation-aware spatial information into prompt-based models like SAM for effective instance segmentation in complex aerial imagery.

4) Limitations and Future Directions: Despite the promising results of our approach, certain limitations remain. First, our method prioritizes mask accuracy, and the multistage cascade refinement introduces additional computational overhead. While it achieves a favorable trade-off between accuracy and speed, applications with strict latency requirements may require faster models. In densely populated remote sensing imagery, the computational advantage of one-stage instance segmentation methods often diminishes. Two-stage methods, such as mask R-CNN, extract fixed-size RoI features for each proposal and predict masks only on these small feature crops. In contrast, one-stage approaches predict masks across full-resolution feature maps or dense grids, often combined with dynamic mask generation techniques (e.g., prototype combination or conditional convolutions), which substantially increase computational cost and memory usage. As a result, one-stage models may exhibit lower FPS than two-stage models in these scenarios (Table IV), highlighting the ongoing challenge of designing efficient instance segmentation models for densely populated remote sensing imagery. Second, although OBB-guided SAM improves semisupervised segmentation overall, its performance still lags behind fully supervised methods for certain fine-grained categories. A representative failure occurs in the airplane category of iSAID, where airplanes often cluster densely and exhibit nearly square aspect ratios. In these cases, the OBB—primarily defined by the fuselage and wings—encompasses substantial background, providing no more precise representation than the HBB. When used in OBB-guided SAM, the minimum enclosing rectangle of the OBB further enlarges the effective region, introducing background clutter into the prompt. Even with OBB-derived keypoints, this expanded region may mislead SAM and degrade mask quality. This suggests that, for complex fine-grained structures with near-unity aspect ratios at large angles, the benefit of OBB guidance can diminish. Future work will focus on improving segmentation for such categories by learning keypoints as more precise prompts for SAM, thereby reducing background interference and enhancing mask quality.

V. Conclusion

In this work, we propose an AFRNet for robust rotated object detection and instance segmentation in remote sensing imagery. Our framework integrates a DARPN to generate orientation-aware proposals via shared modulated deformable convolutions, and a cascade of refinement stages that leverage RRoI Align and Gaussian modeling to enhance feature alignment and stabilize angle regression. Extensive experiments across four benchmark datasets (HRSC2016, SCCOS, DOTA, and DIOR-R) demonstrate that AFRNet achieves state-of-the-art performance in rotated object detection, particularly under strict localization metrics. For instance segmentation, AFRNet outperforms existing baselines on both the VHR10-COCO and iSAID datasets, showing substantial improvements in categories involving elongated or densely packed objects. We also introduce an OBB-guided SAM framework for semisupervised instance segmentation, which leverages high-quality oriented boxes and orientation-aware prompt construction to effectively guide SAM. The proposed method achieves competitive performance with only rotated box annotations, significantly narrowing the gap to fully supervised models. Ablation studies further validate the individual contributions of DARPN, Gaussian-based loss, and cascade refinement, highlighting the efficiency–accuracy trade-off and design effectiveness. Overall, AFRNet provides a unified and extensible solution for rotation-aware detection and segmentation in remote sensing, offering both high accuracy and practical efficiency.

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